

SOC DETECTION & INVESTIGATION LAB

Windows 11 • Sysmon • Splunk Enterprise • Wireshark • MITRE ATT&CK

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Document Control	Value
Document	SOC Detection & Investigation Lab
Version	FINAL v1.2
Environment	CYBER-LAB - Windows 11 virtual machine
Classification	Controlled cybersecurity training / portfolio project
Primary telemetry	Microsoft Sysmon Operational logs
Analytics	Splunk Enterprise
Network analysis	Wireshark
Evidence inventory	34 screenshots across all upload batches

CONTROLLED LAB NOTICE

All suspicious-looking activity was intentionally generated in an isolated training environment. The report demonstrates detection and investigation methodology and does not represent a real compromise.

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PORTFOLIO POSITIONING

This document is structured as a SOC investigation case study rather than a tool walkthrough. It prioritises evidence, correlation, analyst judgement, reproducibility and defensible scoping.

INVESTIGATION LENS

Endpoint telemetry | Process ancestry | DNS/network context | ATT&CK mapping | Incident reporting

1. Executive Summary / Incident Overview

This project demonstrates an end-to-end Security Operations Center (SOC) workflow on a Windows 11 endpoint. Microsoft Sysmon generated high-value endpoint telemetry; Splunk Enterprise provided ingestion, detection and event correlation; and Wireshark validated packet-level DNS, HTTP and ICMP activity.

The controlled investigation identified PowerShell and Windows Command Shell execution, current-user discovery, network configuration and connection discovery, PowerShell web/DNS activity, and a controlled TCP/4444 scenario. The analysis moved beyond isolated event searches by reconstructing process ancestry and correlating Sysmon events with ParentProcessId, ProcessId and ProcessGuid.

Executive field	Assessment
Disposition	Expected / controlled lab activity
Production analogue	Would warrant triage if unexpected on a business endpoint
Primary evidence	Sysmon Event IDs 1, 3 and 22 ingested into Splunk
Strongest correlation	PowerShell -> cmd.exe -> whoami.exe; PowerShell process -> DNS query
Analytical outcome	Detection, process correlation, ATT&CK mapping and defensible reporting completed

Key analytical principle

A technique match or suspicious command is not proof of compromise. The report distinguishes observed telemetry from interpretation and explicitly identifies benign test infrastructure.

2. Investigation Scope & Environment

Component	Implementation	Purpose
Endpoint	Windows 11 VM - CYBER-LAB	Controlled source of process and network telemetry
Virtualisation	VMware Workstation	Isolation and recovery snapshots
Endpoint telemetry	Microsoft Sysmon	Process creation, DNS and network visibility
SIEM	Splunk Enterprise	Search, detection, correlation and reporting
Packet analysis	Wireshark	Independent DNS, HTTP and ICMP validation
Framework	MITRE ATT&CK Enterprise	Behavioural mapping and detection context

Scope was intentionally restricted to a single controlled endpoint. example.com was used as a benign demonstration domain. The TCP/4444 scenario used loopback traffic (127.0.0.1), allowing non-standard-port telemetry to be generated without connecting to a malicious external service.

3. Detection and Investigation Methodology

- Validate host connectivity and packet visibility with command-line tests and Wireshark.
- Confirm Sysmon event generation locally and ingest Microsoft-Windows-Sysmon/Operational into Splunk.
- Build reusable Splunk detections for PowerShell execution, system/network discovery, PowerShell spawning CMD, and non-standard-port activity.
- Generate controlled activity and inspect Image, CommandLine, ParentImage, ParentCommandLine, ProcessId, ParentProcessId and ProcessGuid.
- Use time-window, process-tree and ProcessGuid correlation to connect related events.
- Map supported behaviours to MITRE ATT&CK and preserve screenshots as evidence.

3.1 Core telemetry

Sysmon Event ID	Data type	Analytical use
1	Process Creation	Executable, command line, process ancestry, PID and ProcessGuid
3	Network Connection	Process-associated connection and destination port
22	DNS Query	Process-associated name resolution

3.2 Detection Logic & Reproducible SPL

The following searches capture the principal detection and correlation logic used in the lab. They are included as readable text so the investigation can be reproduced without relying solely on screenshots.

Detection / Pivot	Representative SPL
PowerShell process creation	index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "<EventID>1</EventID>" "powershell.exe"
PowerShell -> Windows Command Shell	index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "<EventID>1</EventID>" "cmd.exe" "ParentImage" "powershell.exe"
Discovery activity	index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "<EventID>1</EventID>" ("whoami.exe" OR "ipconfig.exe" OR "netstat.exe")
Controlled TCP/4444 scenario	index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" "<EventID>3</EventID>" "4444"
PowerShell web/DNS pivot	index=* source="WinEventLog:Microsoft-Windows-Sysmon/Operational" ("Invoke-WebRequest" OR "example.com")

Analyst note: these searches are intentionally transparent and lab-focused. In production they should be adapted to the organisation's field extractions, baselines, allow-lists and risk model.

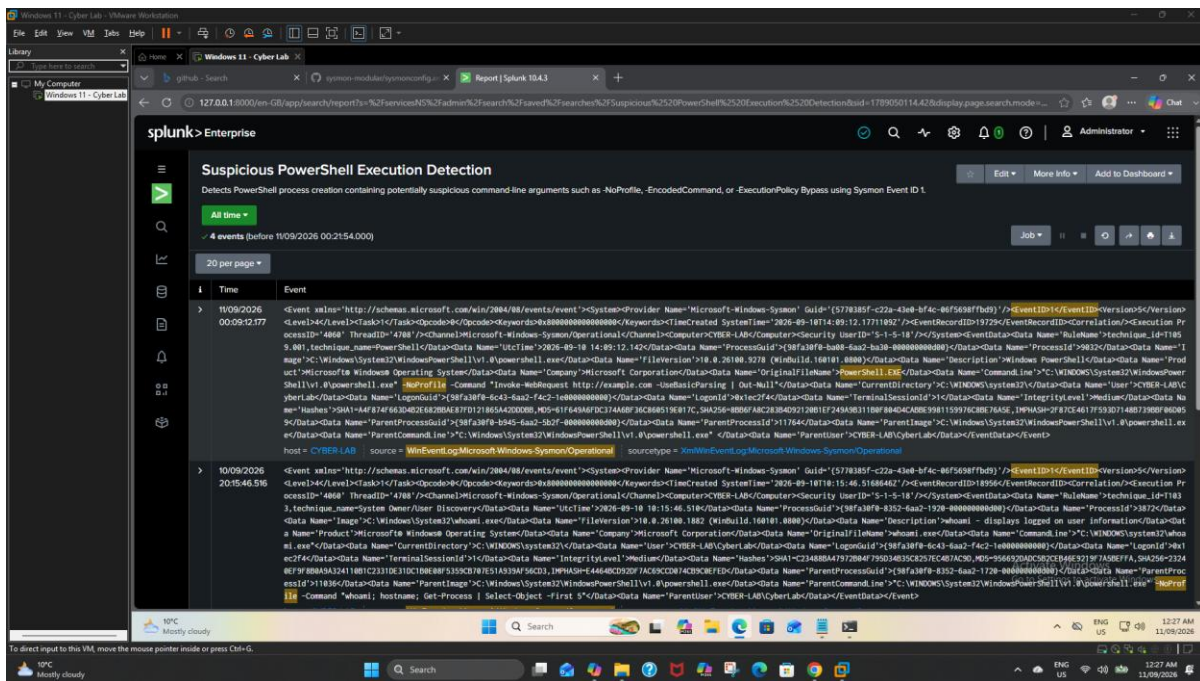
4. Incident Timeline

The timeline below represents the strongest correlated discovery sequence from the controlled incident exercise. The PowerShell web/DNS test and TCP/4444 test were separate controlled scenarios and are analysed in Section 7 rather than being presented as one continuous intrusion.

Time	Observed activity	SOC interpretation
01:22:41	whoami.exe	System owner/user discovery; PowerShell parent PID 5016
01:24:01	ipconfig.exe /all	System network configuration discovery; same PowerShell parent PID 5016
01:24:34	NETSTAT.EXE -ano	System network connections discovery; same PowerShell parent PID 5016
01:26:07	cmd.exe /c "echo INCIDENT-LAB-01 && whoami"	PowerShell spawned Windows Command Shell
01:26:07	whoami.exe	cmd.exe child; completes PowerShell -> CMD -> whoami chain

5. Technical Findings & Evidence

Sysmon Event ID 1 and Splunk searches identified PowerShell process creation and command-line activity. PowerShell is a legitimate administrative platform; therefore, execution alone is not treated as malicious. Its value comes from command-line context, parent/child relationships and associated network activity.



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[illegible]

5.3 PowerShell spawning Windows Command Shell

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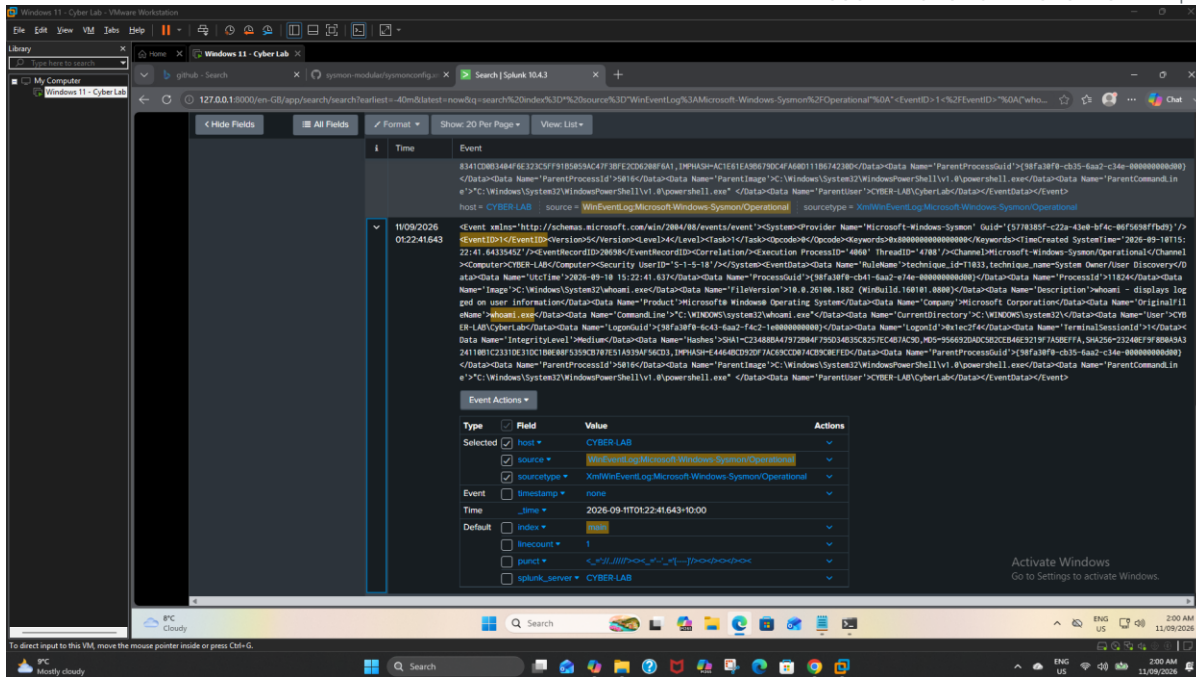


Figure 6. ParentProcessId evidence supporting common PowerShell ancestry.

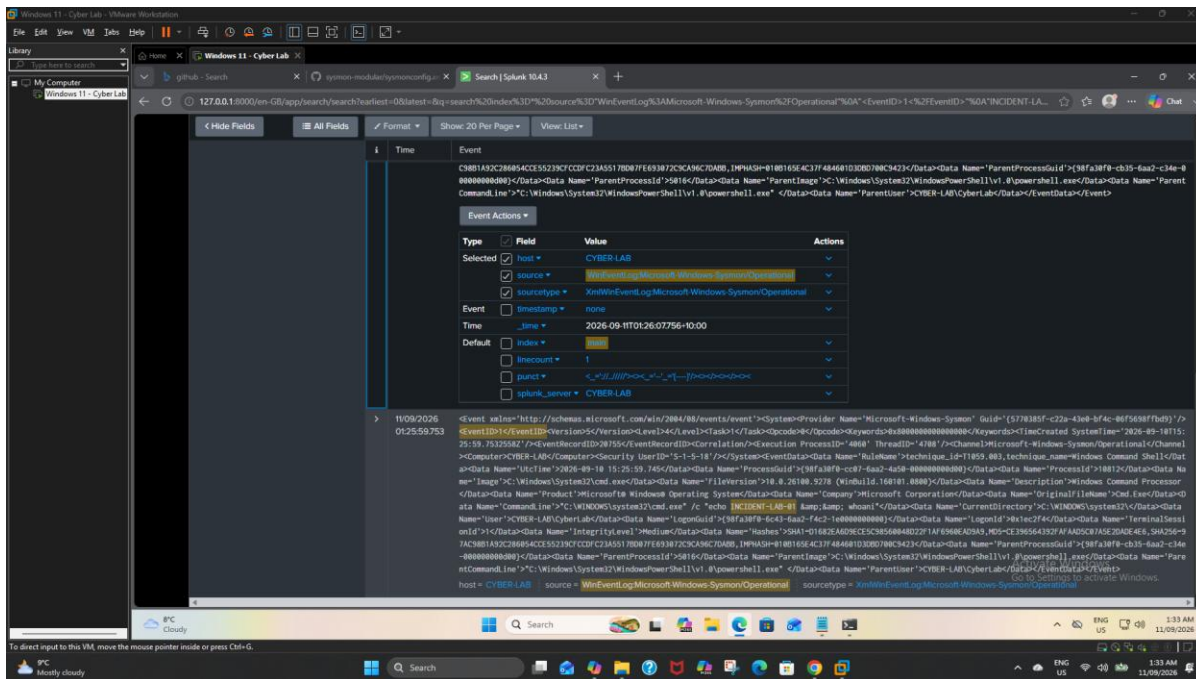


Figure 7. Raw process-chain evidence supporting PowerShell -> cmd.exe -> whoami.exe.

7. Network / DNS Analysis

7.1 Packet-level validation

Wireshark independently validated DNS, HTTP and ICMP visibility. This provided a network perspective alongside endpoint telemetry and demonstrated the ability to interpret both packet-level and host-level evidence.

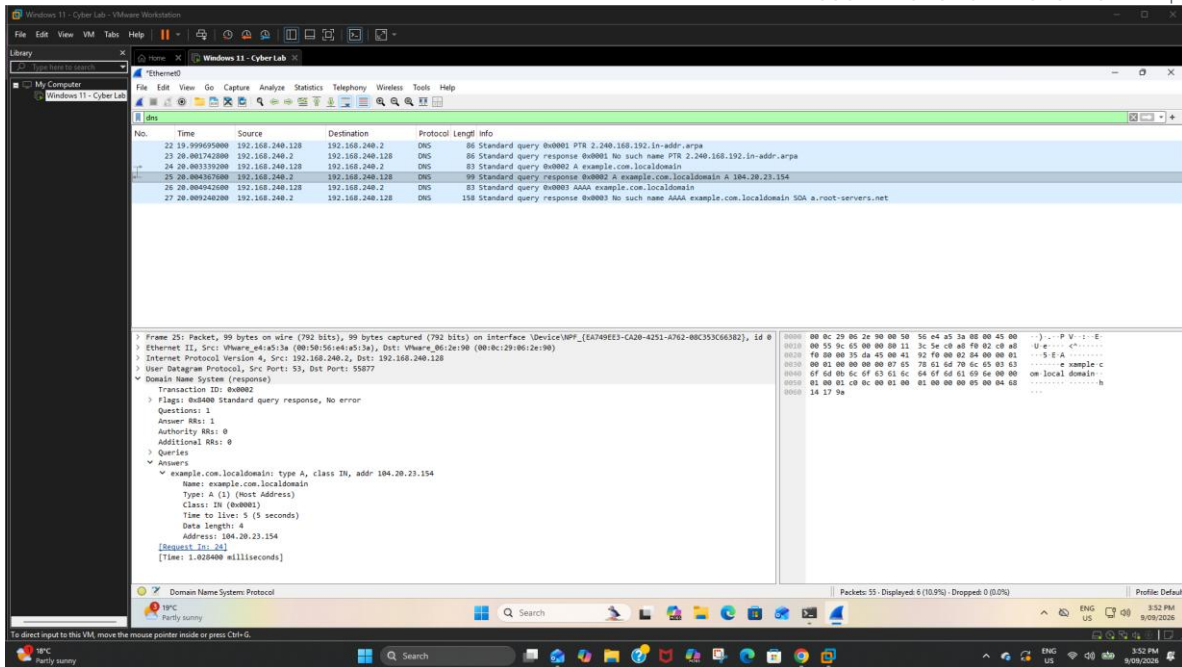


Figure 8. Wireshark DNS analysis from the controlled lab.

7.2 PowerShell web and DNS correlation

A controlled PowerShell Invoke-WebRequest request to <http://example.com> generated process-creation and DNS telemetry. The investigation extracted the PowerShell ProcessGuid/ProcessId and pivoted across Sysmon events to associate that exact process instance with Event ID 22 DNS activity for example.com.

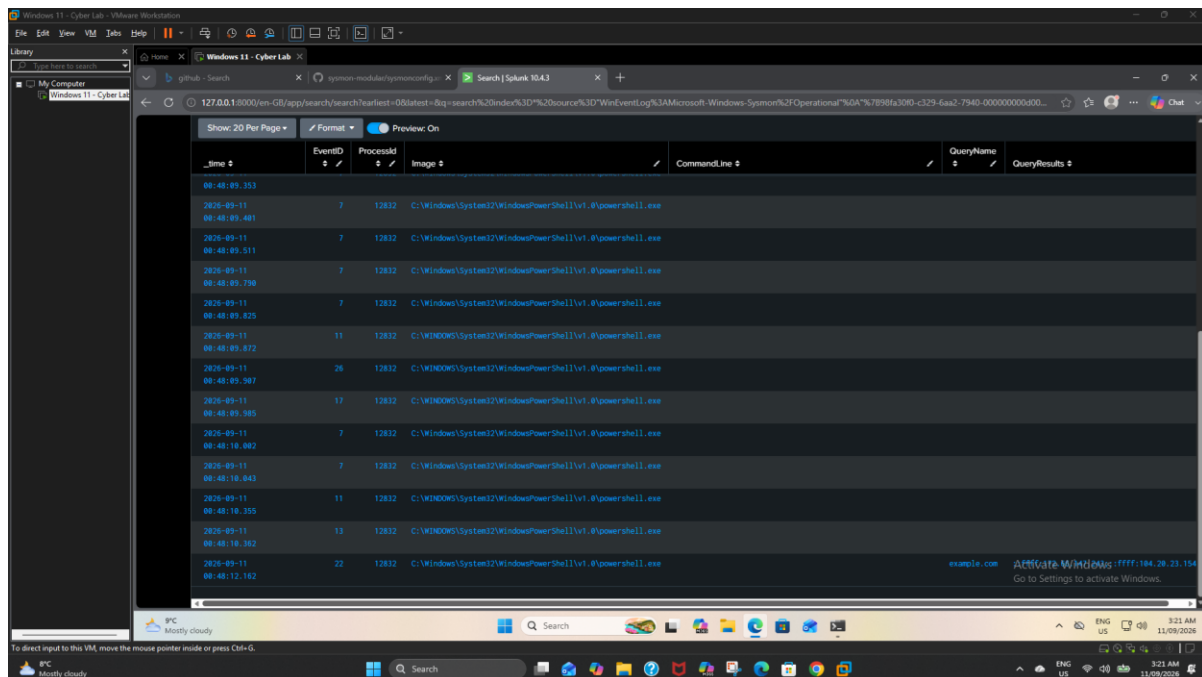


Figure 9. ProcessGuid/ProcessId correlation linking PowerShell execution and DNS activity.

7.3 TCP/4444 context

Port 4444 is often scrutinised, but a port number alone does not establish MITRE ATT&CK T1571. T1571 specifically concerns protocol/port pairings that are atypical for the protocol. In this lab, the TCP/4444 loopback activity is retained as a controlled detection scenario aligned to the concept of non-standard-port monitoring, not as proof of C2.

8. MITRE ATT&CK Mapping

The mapping below was quality-checked against the current MITRE ATT&CK Enterprise technique pages. The report distinguishes direct behavioural mappings from a lab scenario that merely demonstrates a related detection concept.

Technique	Name	Evidence	Assessment
T1059.001	PowerShell	powershell.exe process creation and command-line activity	Direct behavioural mapping
T1059.003	Windows Command Shell	cmd.exe spawned from PowerShell	Direct behavioural mapping
T1033	System Owner/User Discovery	whoami.exe	Direct behavioural mapping
T1016	System Network Configuration Discovery	ipconfig.exe /all	Direct behavioural mapping
T1049	System Network Connections Discovery	netstat.exe -ano	Direct behavioural mapping
T1571	Non-Standard Port	Controlled TCP/4444 loopback scenario	Detection scenario / concept alignment; not proof of C2

ATT&CK accuracy note

MITRE currently identifies whoami under T1033, ipconfig under T1016, and netstat under T1049. T1571 requires a non-standard protocol/port pairing; the lab screenshot is therefore described conservatively as a detection scenario.

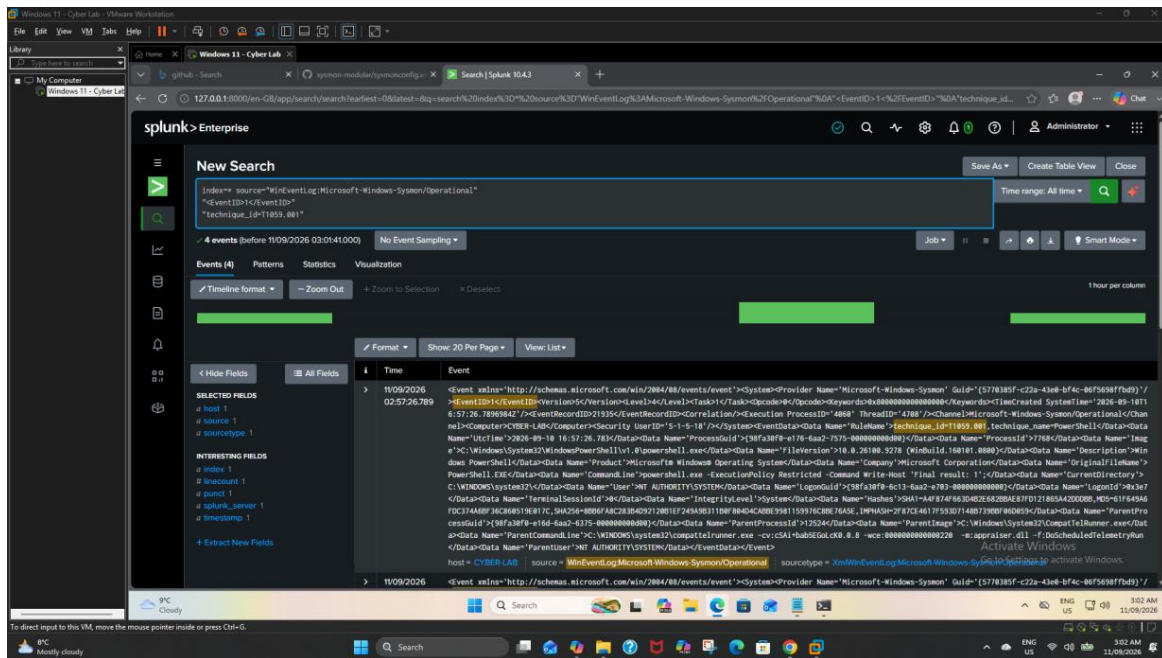


Figure 10. T1059.001 - PowerShell evidence.

The screenshot displays the Splunk Enterprise web interface. At the top, there's a navigation bar with icons for home, search, and other functions. Below this, a sidebar on the left contains navigation icons. The main content area is titled 'New Search' and shows a search query: `index=* source="WinEventLog:Microsoft-Windows-System/Operational" "EventID:4" EventID="technique_id:1" | ipconfig.exe`. The search results are displayed in a table with columns for Time and Event. The first result is from 11/09/2026 at 01:24:01.391, showing a system configuration change. The interface also includes a sidebar with navigation icons, a top navigation bar, and a bottom status bar.

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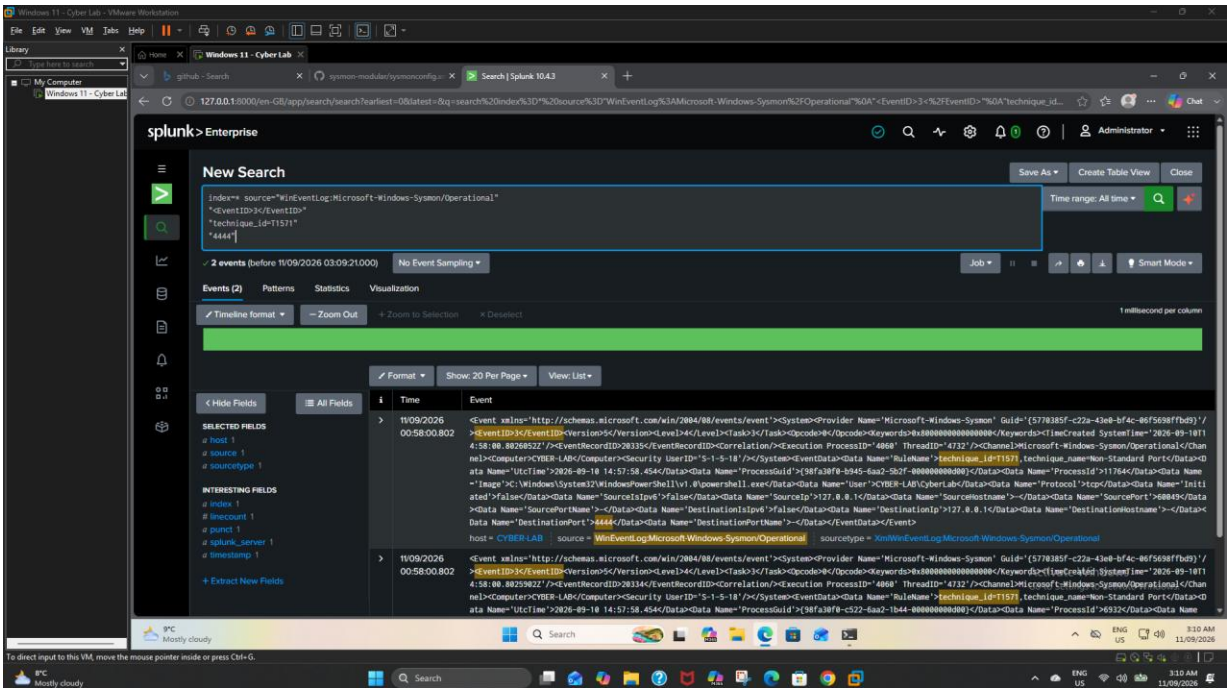


Figure 13. Lab rule labelled T1571; interpreted conservatively in this report.

8.1 Reference basis

- MITRE ATT&CK T1059.001 - Command and Scripting Interpreter: PowerShell.
- MITRE ATT&CK T1059.003 - Command and Scripting Interpreter: Windows Command Shell.
- MITRE ATT&CK T1033 - System Owner/User Discovery (includes whoami).
- MITRE ATT&CK T1016 - System Network Configuration Discovery (includes ipconfig).
- MITRE ATT&CK T1049 - System Network Connections Discovery (includes netstat).
- MITRE ATT&CK T1571 - Non-Standard Port.

9. Indicators of Compromise / Observables

Because this is a controlled lab, these items are more accurately described as observables and investigation pivots rather than confirmed Indicators of Compromise (IOCs).

Type	Observable	Context
Host	CYBER-LAB	Controlled Windows 11 endpoint
User	CYBER-LAB\CyberLab	Lab account context
Processes	powershell.exe; cmd.exe; whoami.exe; ipconfig.exe; NETSTAT.EXE	Execution and discovery pivots
Marker	INCIDENT-LAB-01	Controlled incident correlation marker
Domain	example.com	Benign demonstration domain
Port	TCP/4444	Controlled loopback test
Destination	127.0.0.1	Loopback address in port scenario
Log source	Microsoft-Windows-Sysmon/Operational	Primary endpoint evidence source

Process IDs and ProcessGuids shown in screenshots are ephemeral correlation identifiers. They are useful inside the investigation but should not be treated as durable threat indicators.

10. Analyst Assessment

Assessment field	Conclusion
Incident status	Controlled lab activity - no real compromise
Evidence confidence	High for process ancestry and DNS/process correlation
Maliciousness confidence	Not applicable; activity was intentionally generated
Production concern	Moderate-to-high triage priority if the same clustered behaviour were unexpected
Primary rationale	Scripting + shell execution + rapid discovery + unusual network context merits investigation

The strongest part of the investigation is not any single command; it is the correlation. Parent-child relationships and ProcessGuid/ProcessId pivots demonstrate that related events can be tied to the same execution context. This reduces the risk of constructing an incident narrative from unrelated events that merely occurred near each other in time.

In production, severity would depend on user intent, asset criticality, destination reputation, persistence, credential access, lateral movement and corroborating telemetry. Dual-use administrative commands should not be classified as malicious without that context.

10.1 Investigation Limitations & Assumptions

- Single Windows 11 virtual endpoint; findings are not representative of an enterprise-wide hunt.
- Telemetry scope centred on Sysmon Operational events ingested into Splunk, supplemented by Wireshark packet captures.
- No production EDR, identity, email, cloud, proxy or threat-intelligence telemetry was available for corroboration.
- All suspicious-looking activity was intentionally generated for training; no real malware or unauthorised compromise was introduced.
- example.com was a benign demonstration destination, and the TCP/4444 test used 127.0.0.1 loopback traffic.
- ATT&CK mappings describe observed behaviour or detection concepts; they do not independently establish malicious intent.

11. Containment & Remediation Recommendations

For an equivalent unexplained event on a production endpoint, a SOC analyst should:

- Validate whether PowerShell, cmd.exe and discovery commands were authorised by the user, administrator or automation platform.
- Review surrounding endpoint, PowerShell, DNS, proxy/firewall and authentication telemetry using the same time window and process ancestry.
- Isolate the endpoint when malicious execution or command-and-control is reasonably suspected and organisational procedure supports containment.
- Preserve relevant evidence before destructive remediation, including command-line, process, network and authentication context.
- Block confirmed malicious domains/IPs and hunt for matching process ancestry, command lines and ATT&CK behaviours across other endpoints.
- Remove persistence or malicious artifacts, patch the root cause, rotate exposed credentials where required, and monitor for recurrence.

12. Conclusion

The SOC Detection & Investigation Lab demonstrates a complete junior-SOC workflow: network validation, Sysmon telemetry collection, Splunk ingestion, detection engineering, process and DNS correlation, ATT&CK mapping, analyst assessment and evidence-based reporting. The project shows the ability to move from raw Windows telemetry to a defensible investigation narrative rather than relying on isolated alerts.

12.1 Skills demonstrated

- Windows 11 / VMware lab administration and recovery snapshots
- Wireshark DNS, HTTP, ICMP and traceroute analysis
- Sysmon process, DNS and network telemetry interpretation
- Splunk Enterprise ingestion, SPL searching and reusable detection reports
- Detection engineering for PowerShell, discovery, process ancestry and network activity
- Parent-child process analysis, PID correlation and Sysmon ProcessGuid correlation
- MITRE ATT&CK behavioural mapping with conservative analytical interpretation
- SOC incident reconstruction, evidence handling and professional reporting

Portfolio takeaway

This project demonstrates practical SOC analysis: telemetry collection, detection logic, correlation, ATT&CK mapping, evidence interpretation and incident reporting.

12.2 Technical References

- MITRE ATT&CK. Enterprise techniques used in this report: T1059.001 PowerShell; T1059.003 Windows Command Shell; T1033 System Owner/User Discovery; T1016 System Network Configuration Discovery; T1049 System Network Connections Discovery; T1571 Non-Standard Port. Accessed September 2026.
- Microsoft Learn. Sysmon documentation and Sysinternals Sysmon event schema, including Process Create (Event ID 1), Network Connect (Event ID 3) and DNS Query (Event ID 22). Accessed September 2026.
- Splunk Documentation. Search Processing Language (SPL), search and reporting concepts used for telemetry investigation and correlation. Accessed September 2026.
- Wireshark Documentation. User Guide and display-filter concepts used for packet-level DNS, HTTP and ICMP validation. Accessed September 2026.

Reference note: product and framework names are used for technical attribution. URLs are intentionally omitted from the body so the portfolio report remains clean; authoritative vendor/framework documentation should be consulted for current syntax and technique definitions.

13. Evidence Appendix - Complete Screenshot Set

All 34 screenshots supplied across every upload batch are retained below. The main body uses selected high-value evidence; this appendix preserves the complete workflow for portfolio verification and interview discussion.

Network & Packet Analysis

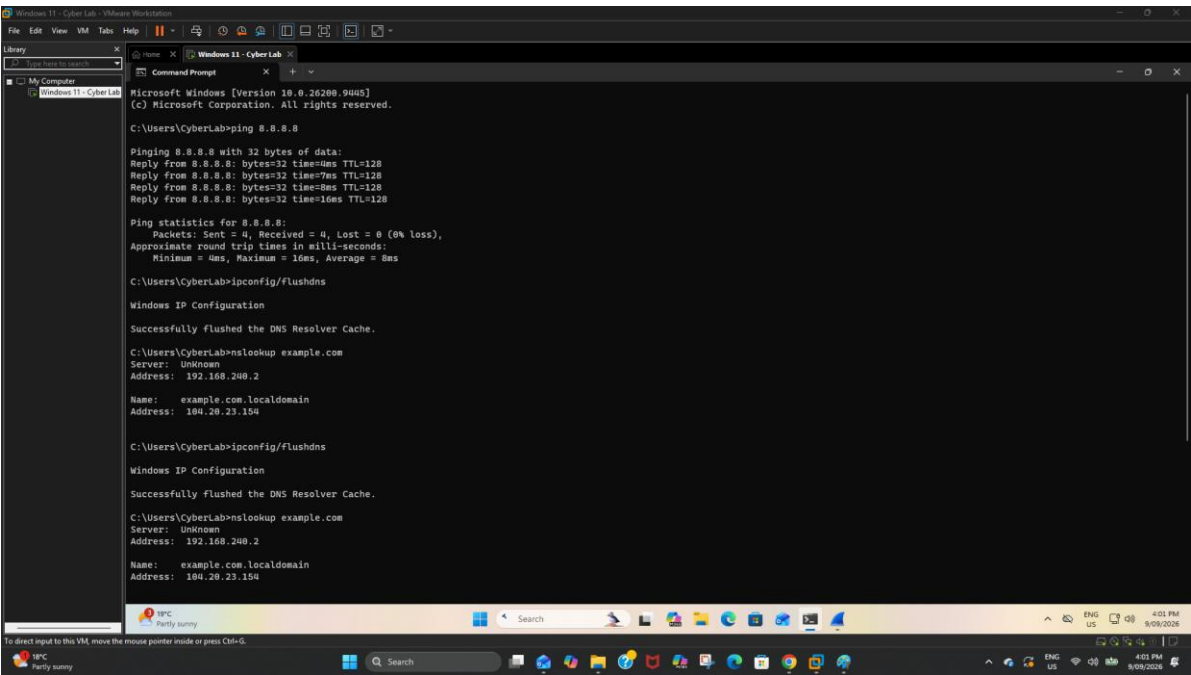


Figure 14. CMD Network Test.

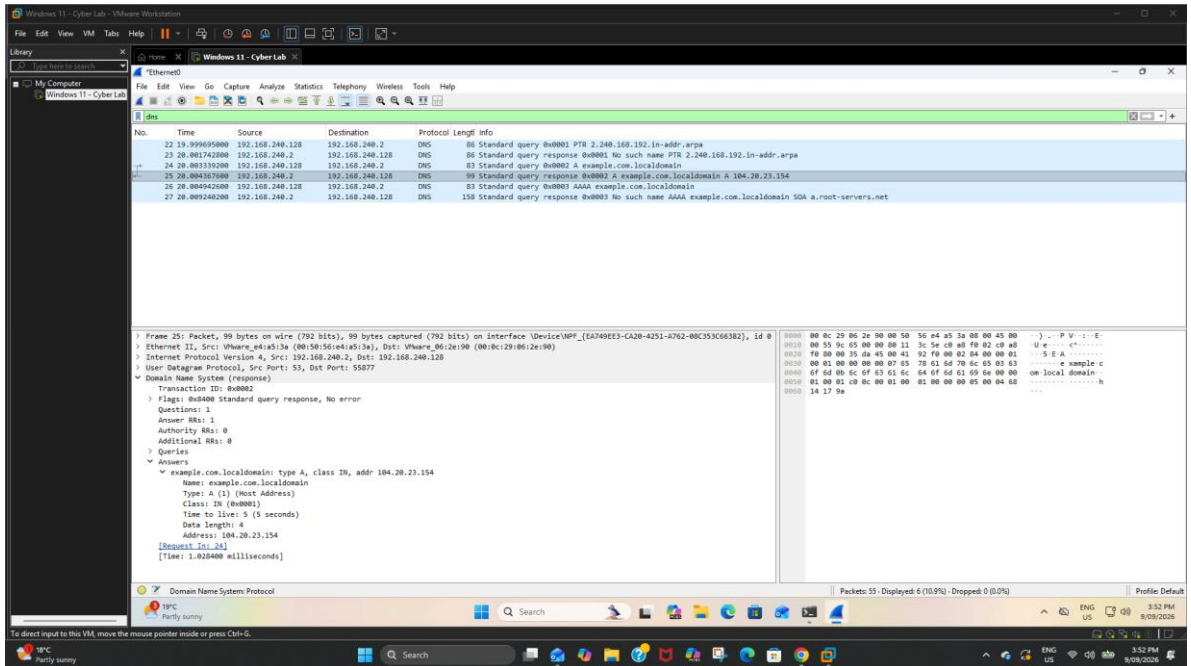


Figure 15. Wireshark DNS Analysis.

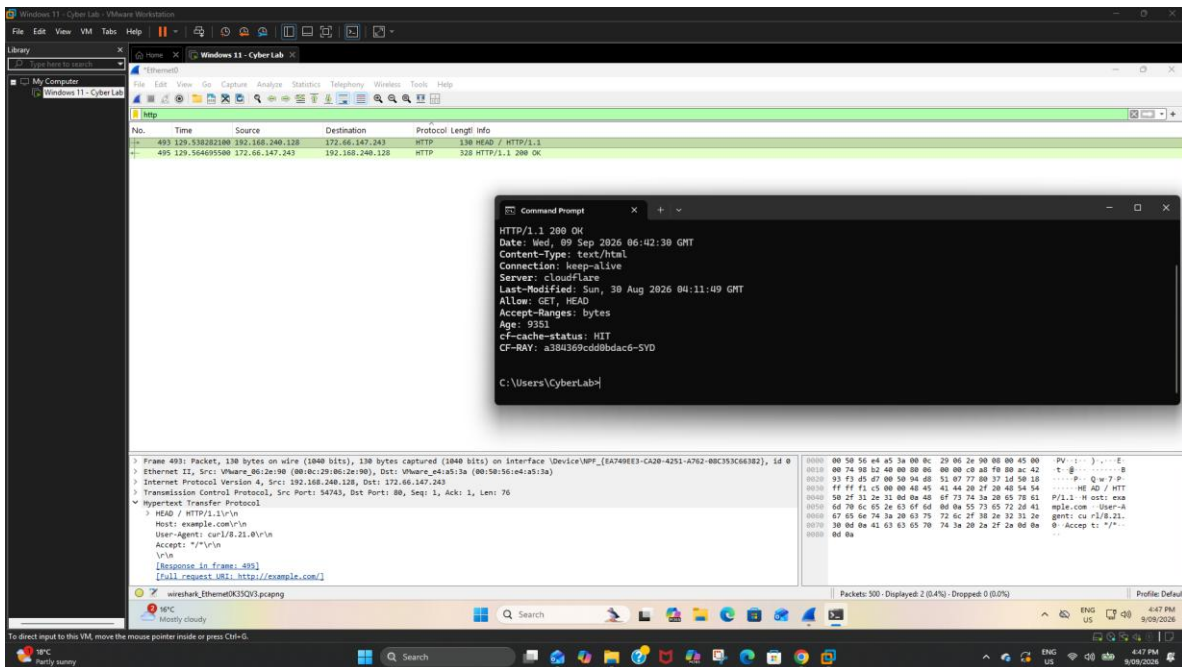


Figure 16. HTTP Request Analysis.

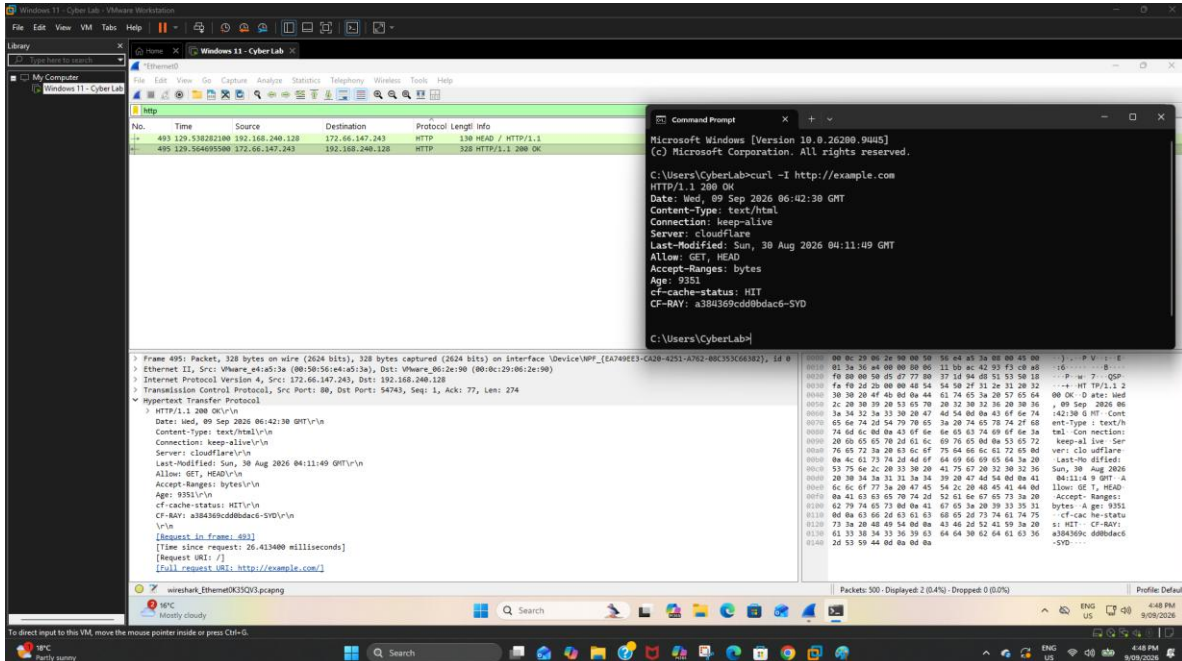
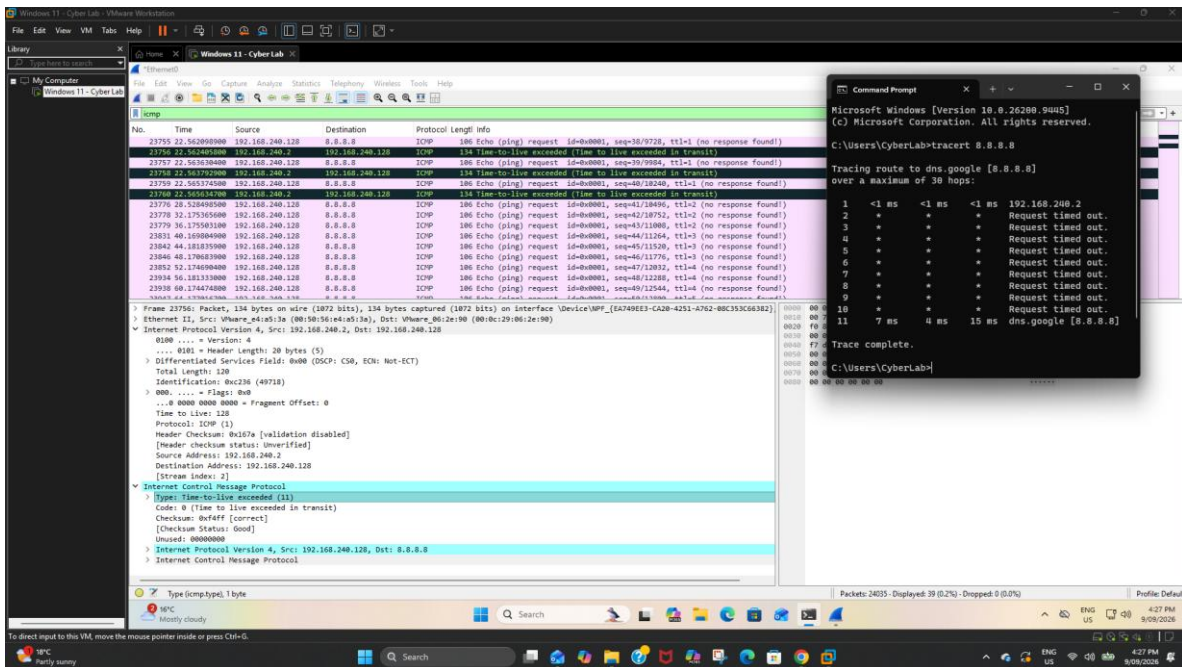
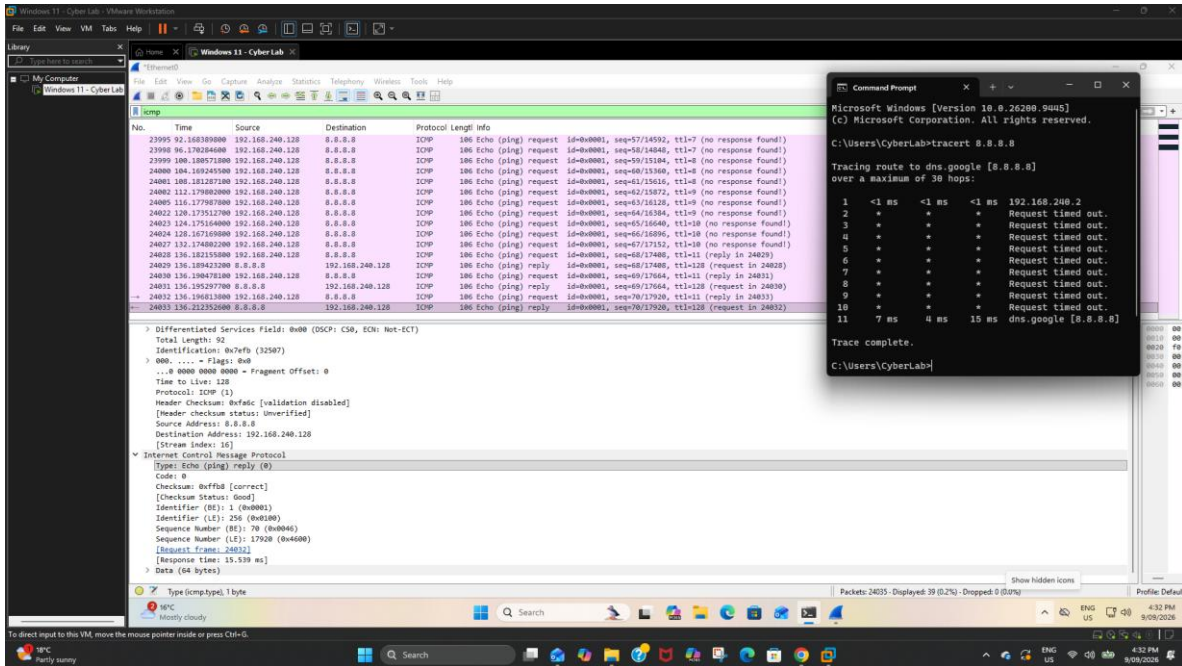


Figure 17. HTTP Response Analysis.



Sysmon Telemetry

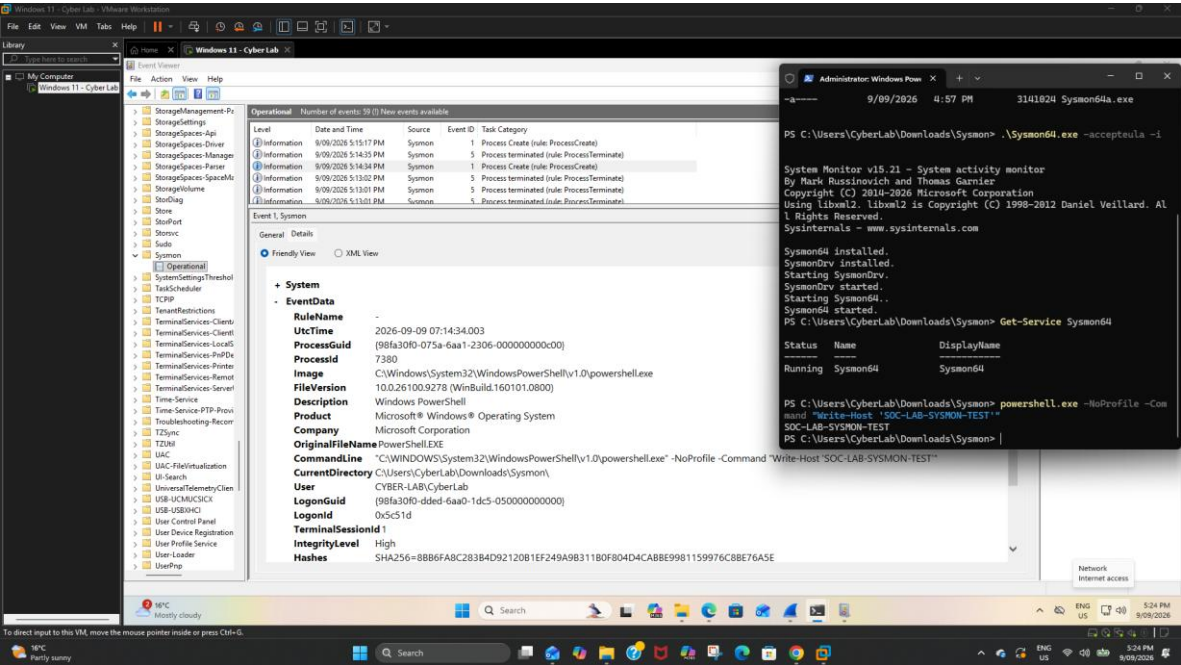


Figure 20. Sysmon PowerShell Detection Evidence.

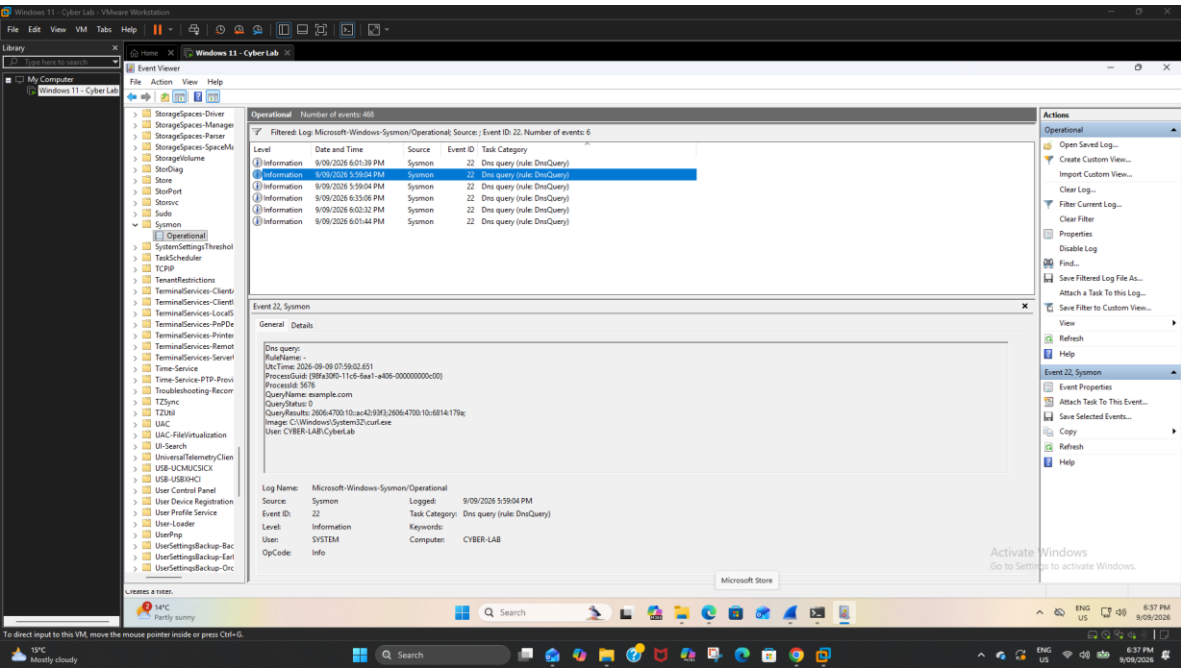


Figure 21. Sysmon Event ID 22 DNS Query.

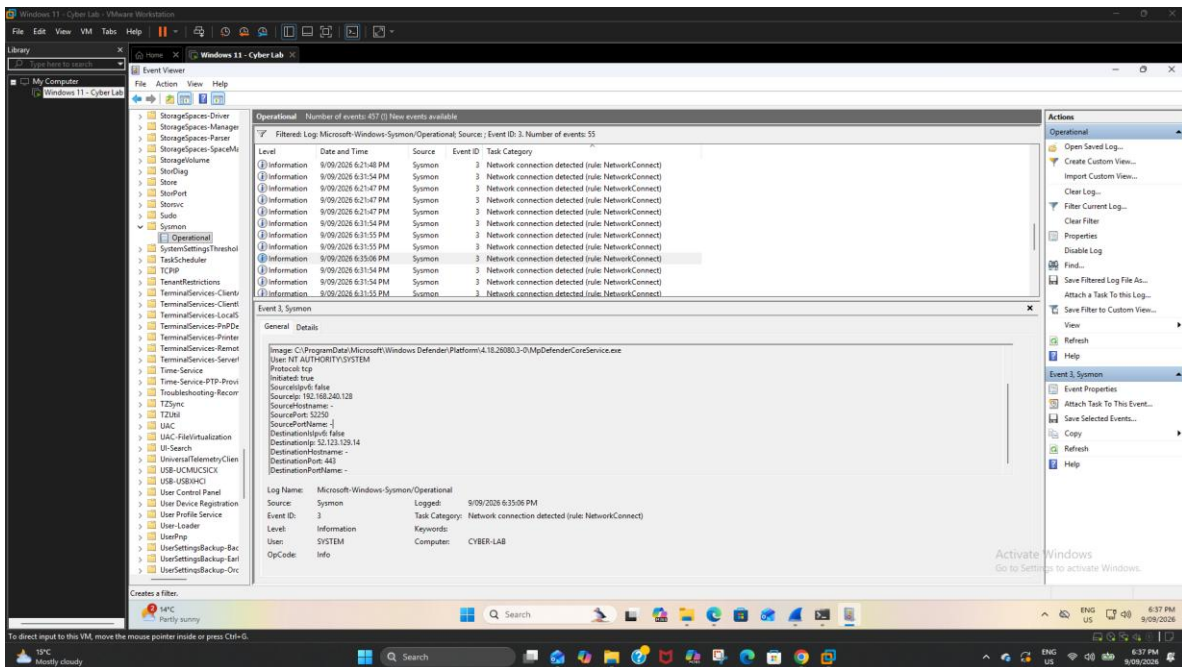


Figure 22. Sysmon Event ID 3 Network Connection.

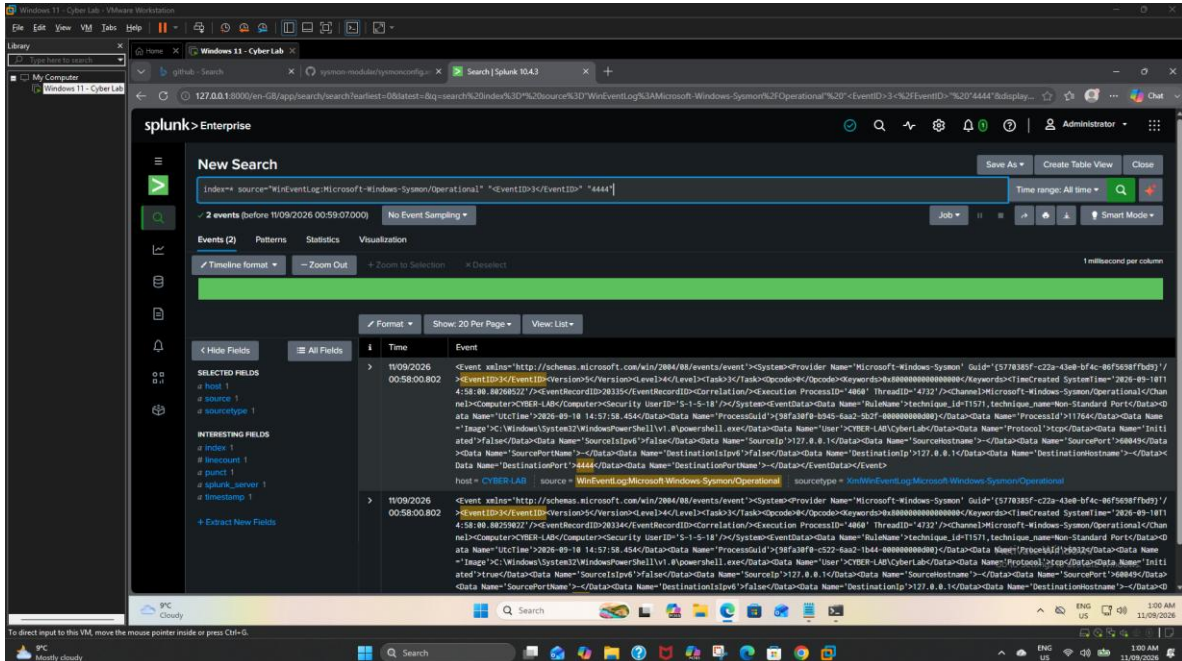


Figure 23. Sysmon Event ID 3 Non-Standard Port 4444.

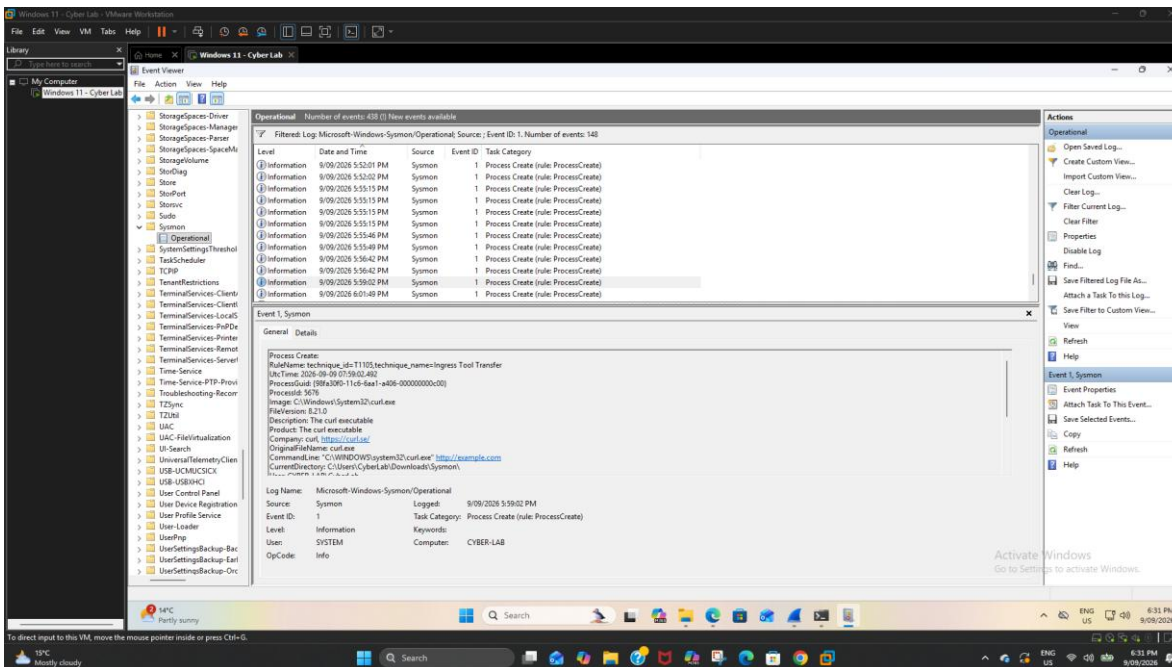


Figure 24. Sysmon MITRE T1105 Curl Detection.

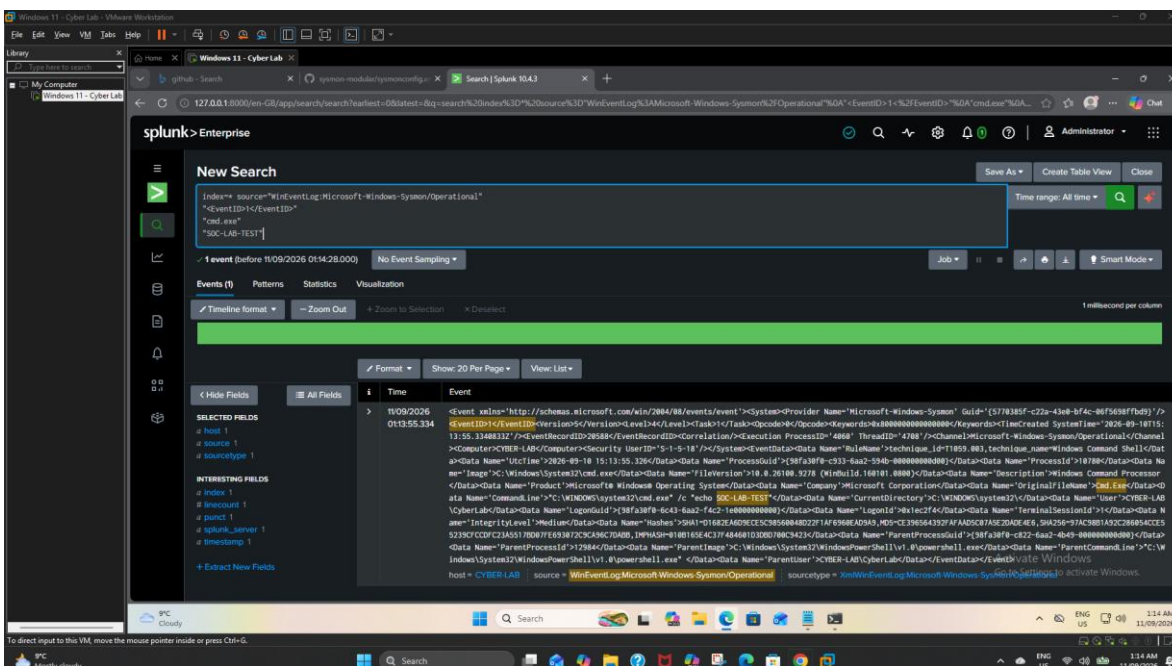


Figure 25. CMD Process Creation - Sysmon Event ID 1

Splunk Detection & Correlation

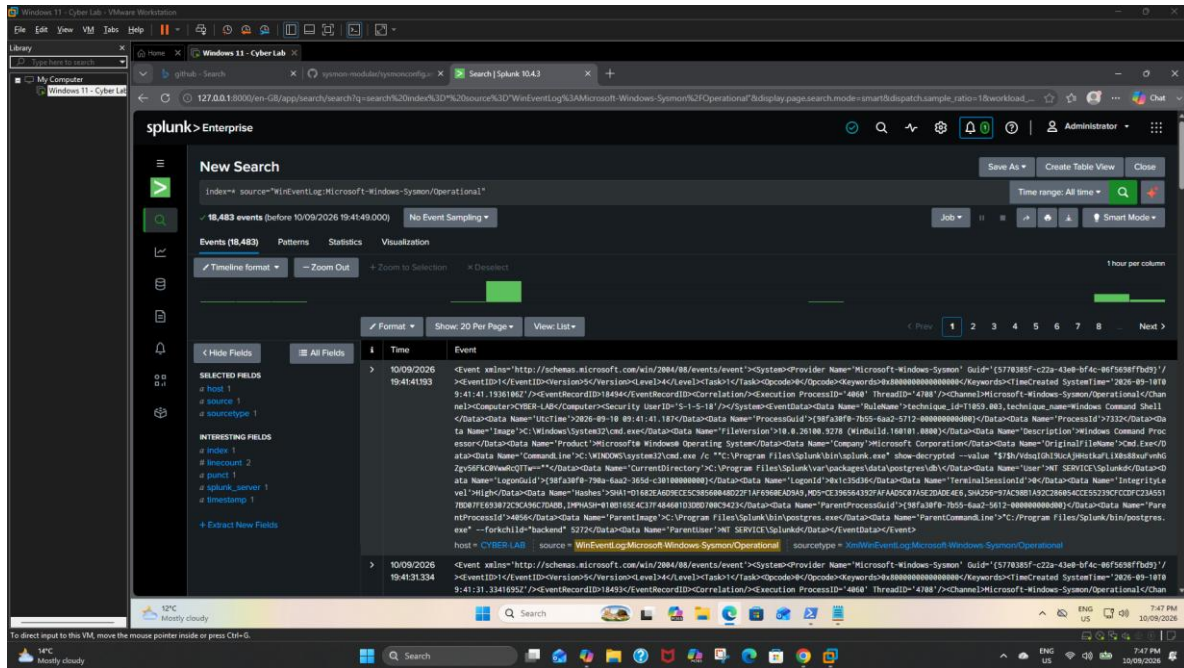


Figure 26. Splunk Sysmon Log Ingestion.

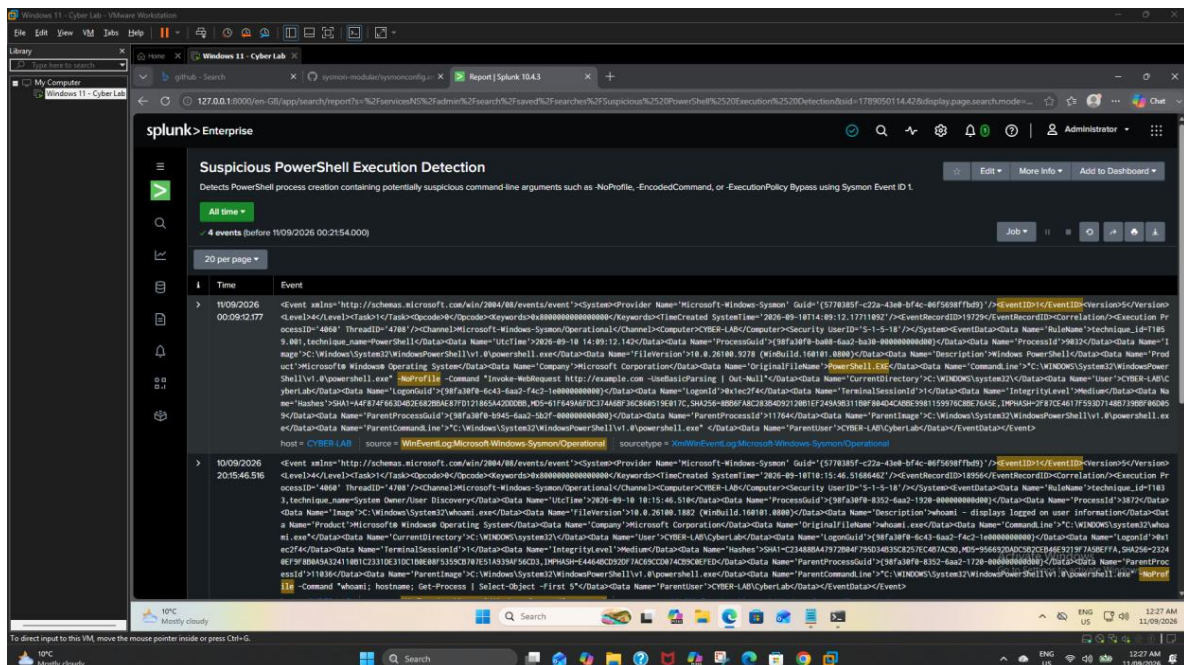


Figure 27. Suspicious PowerShell Detection Report.

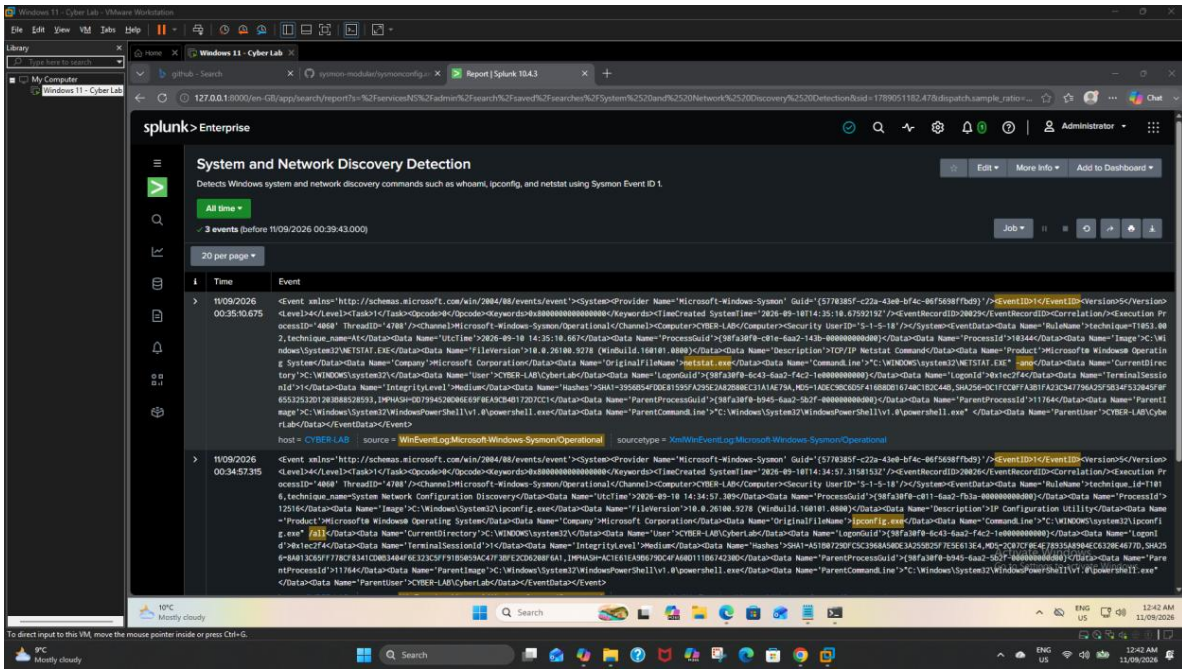


Figure 28. System and Network Discovery Detection.

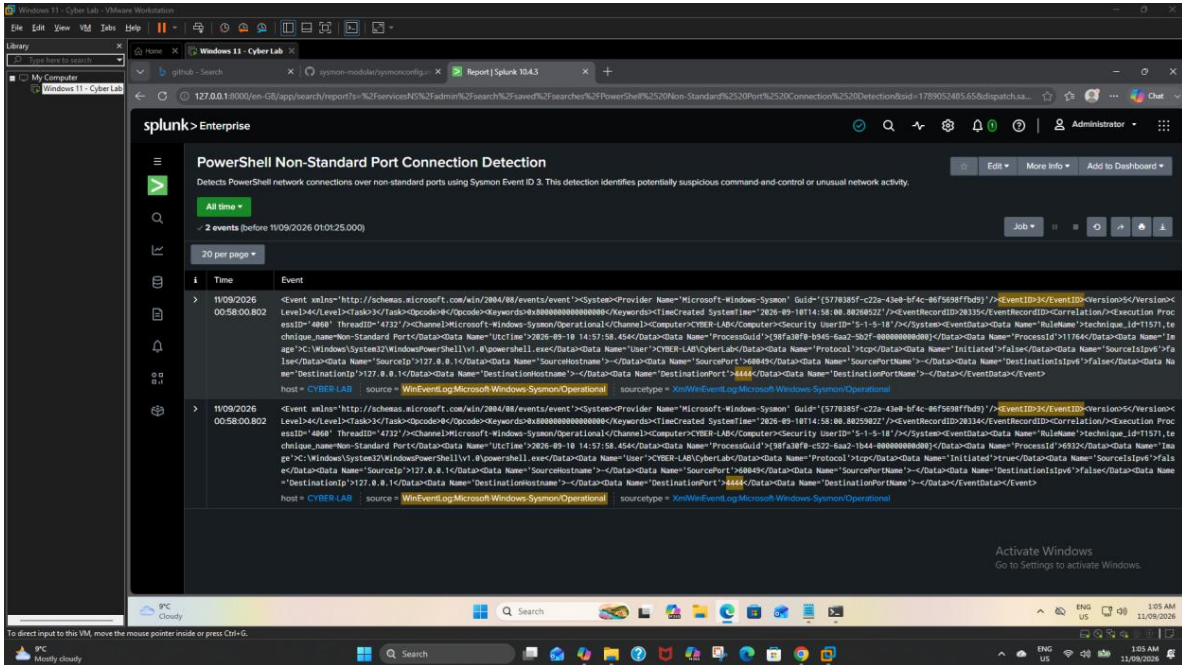


Figure 29. PowerShell Non-Standard Port Detection.

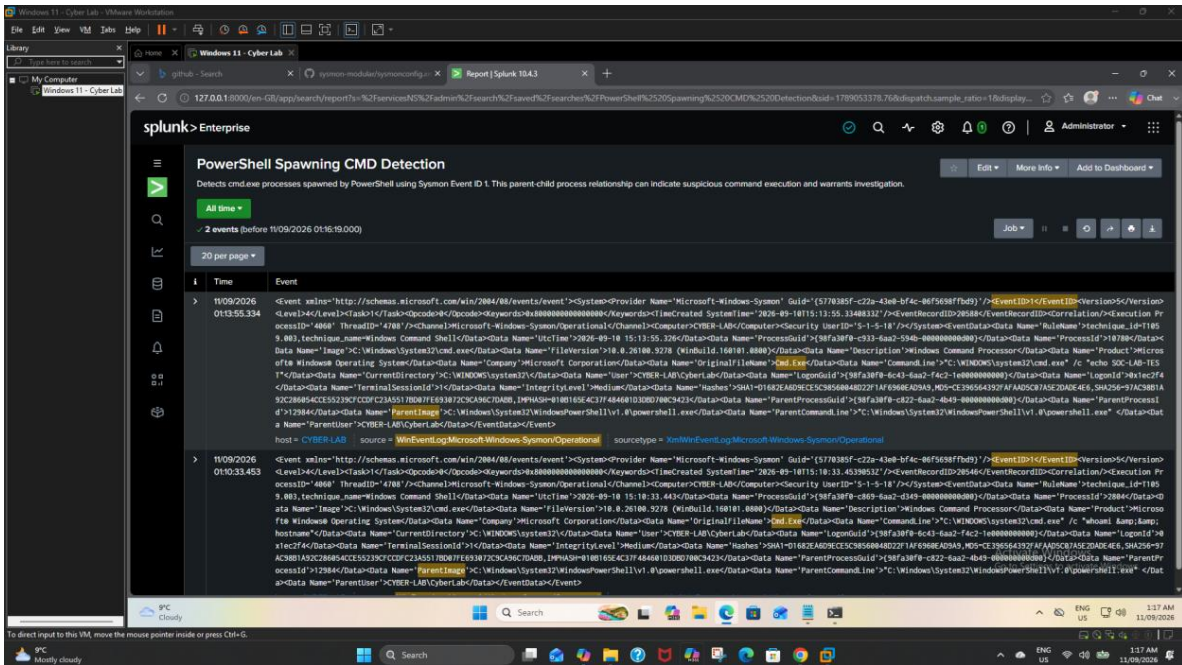


Figure 30. PowerShell Spawning CMD Detection.

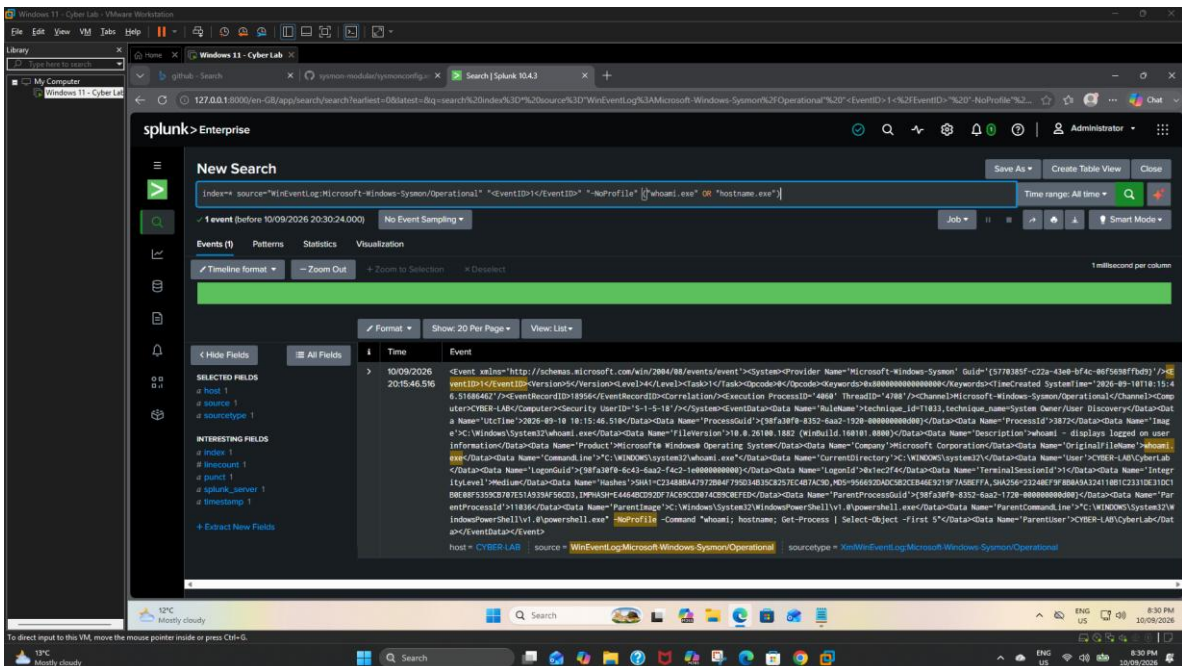


Figure 31. PowerShell Child Process Detection.

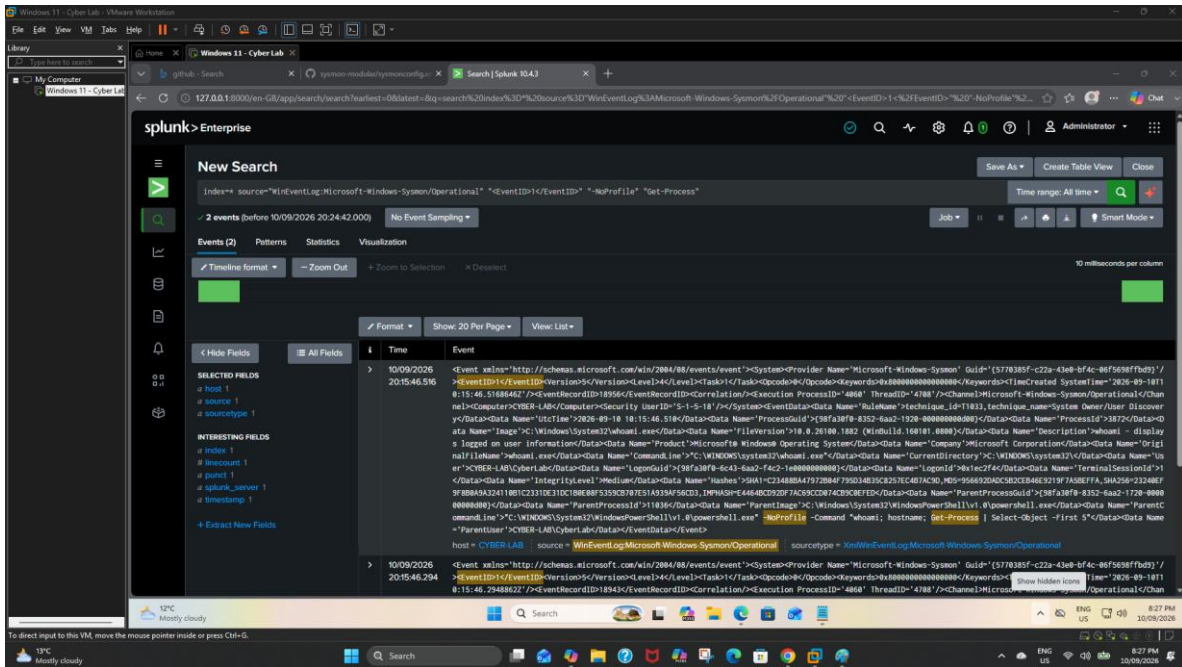


Figure 32. PowerShell Process Investigation.

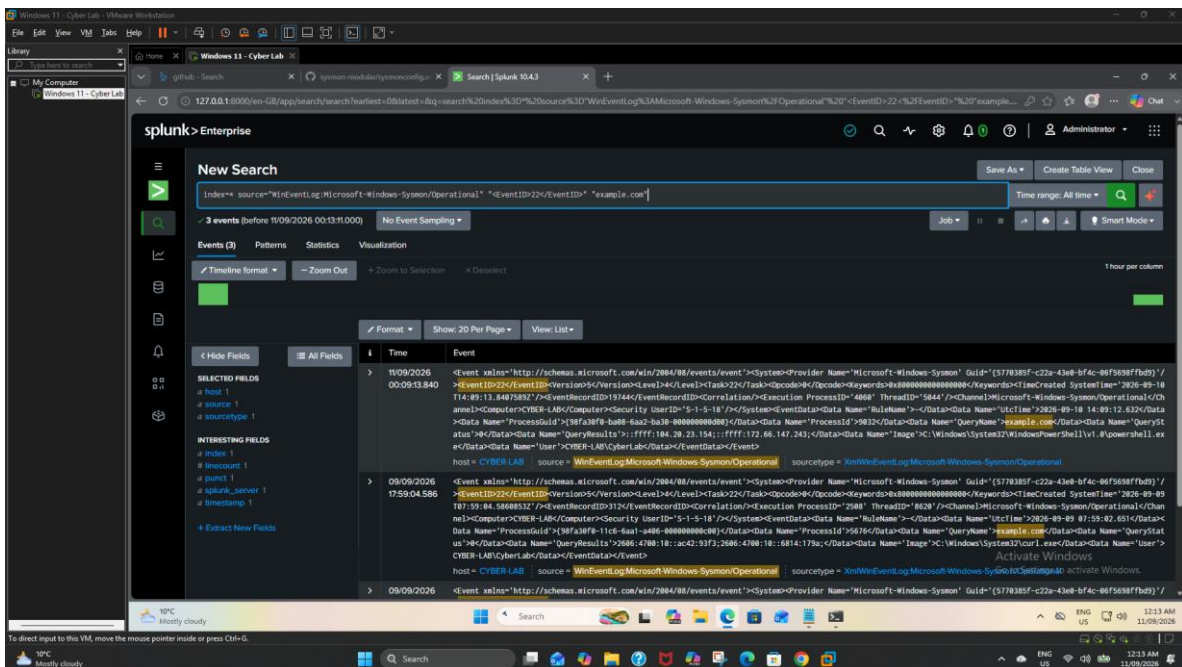


Figure 33. PowerShell DNS Event 22.

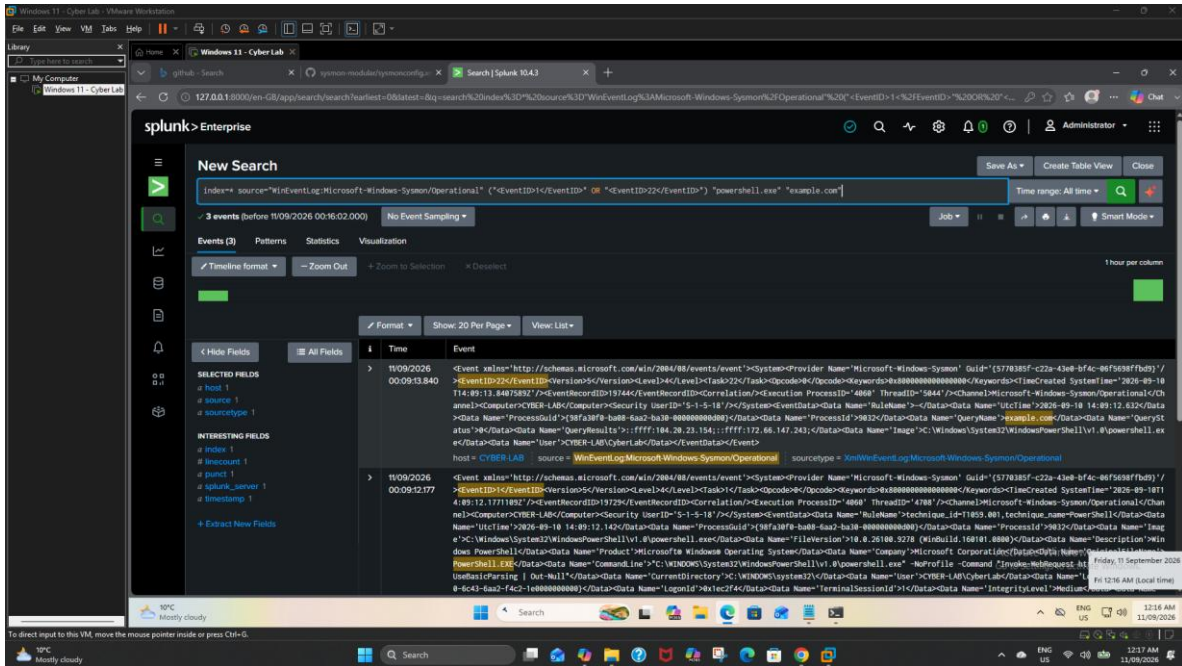


Figure 34. PowerShell DNS Correlation.

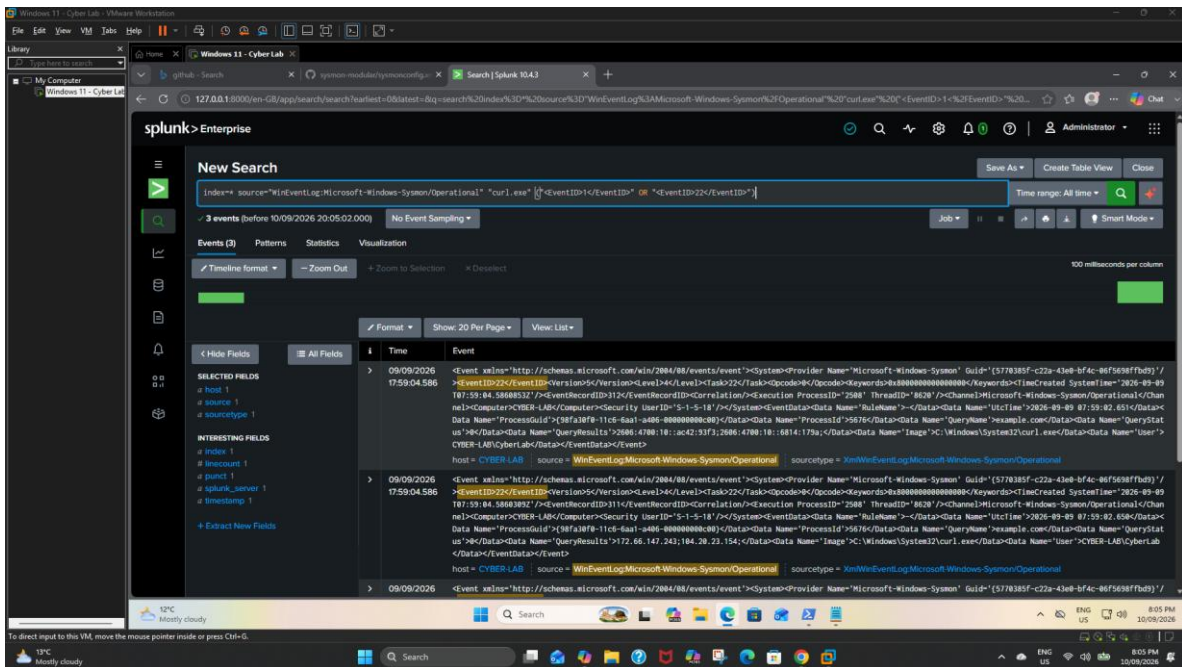


Figure 35. Sysmon Event Correlation.

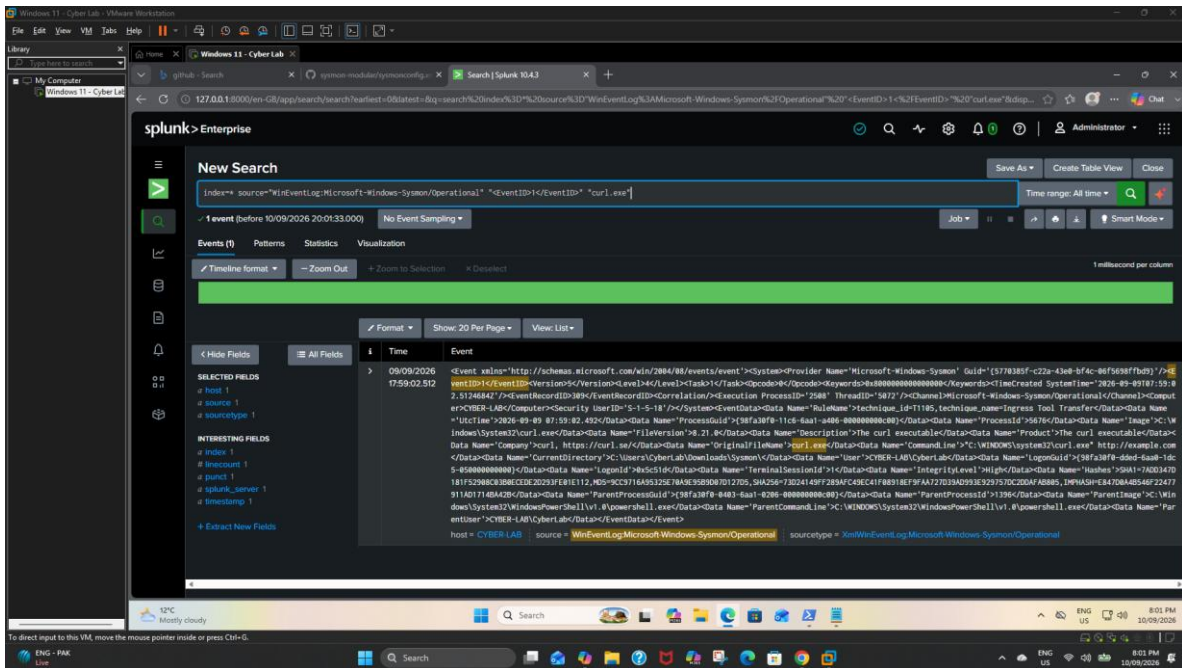


Figure 36. Sysmon Event ID 1 Curl Process.

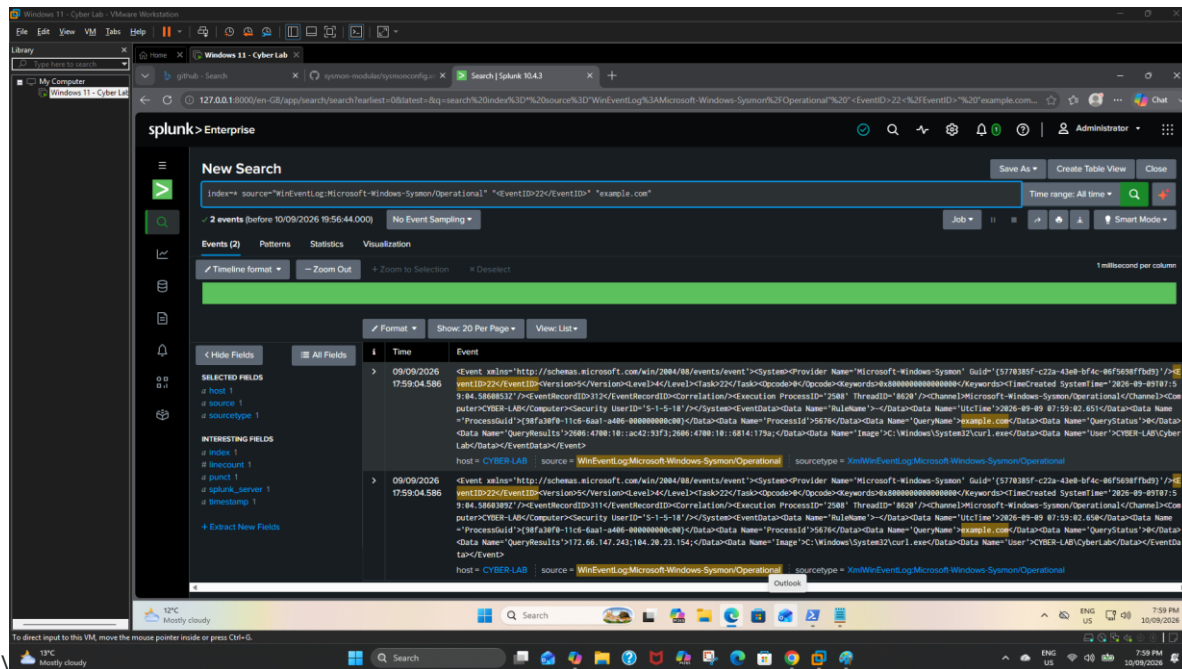


Figure 37. Sysmon DNS Event 22 Curl

Incident Investigation

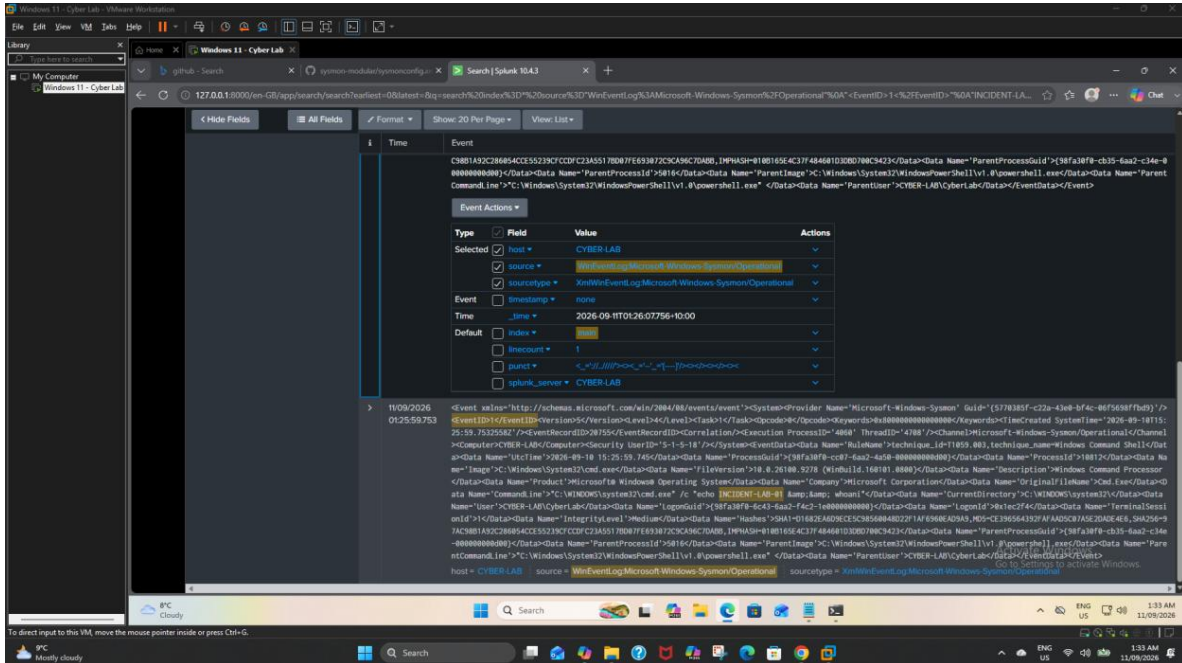


Figure 38. Incident Investigation Process Chain.

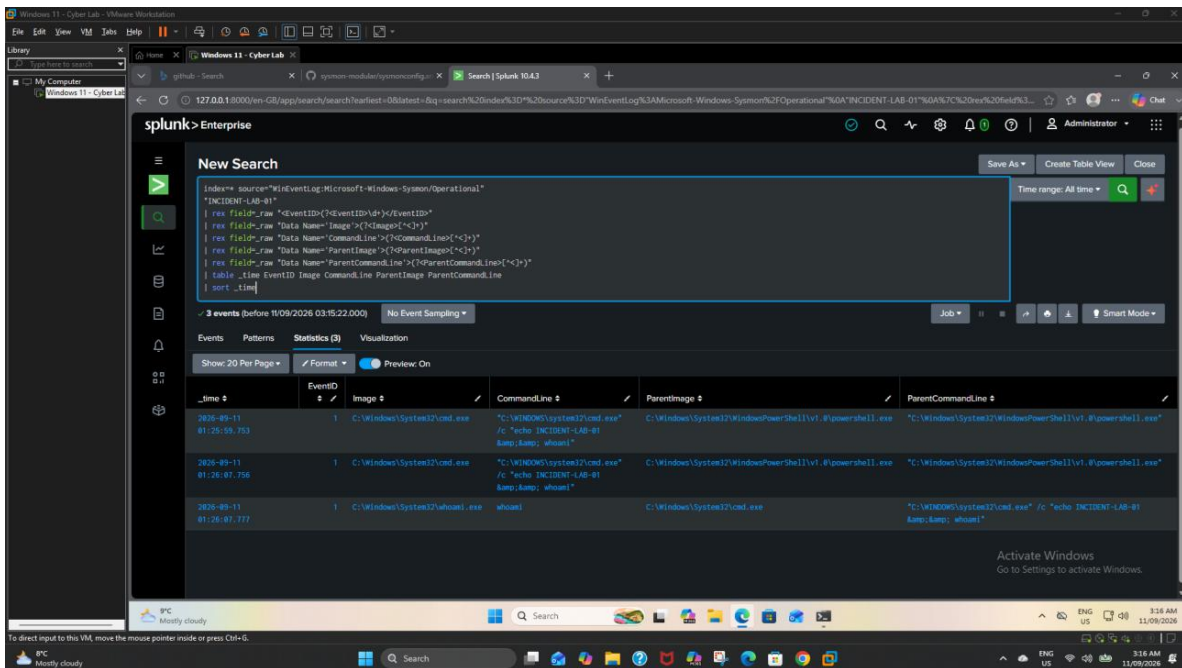


Figure 39. Incident Process Chain Investigation.

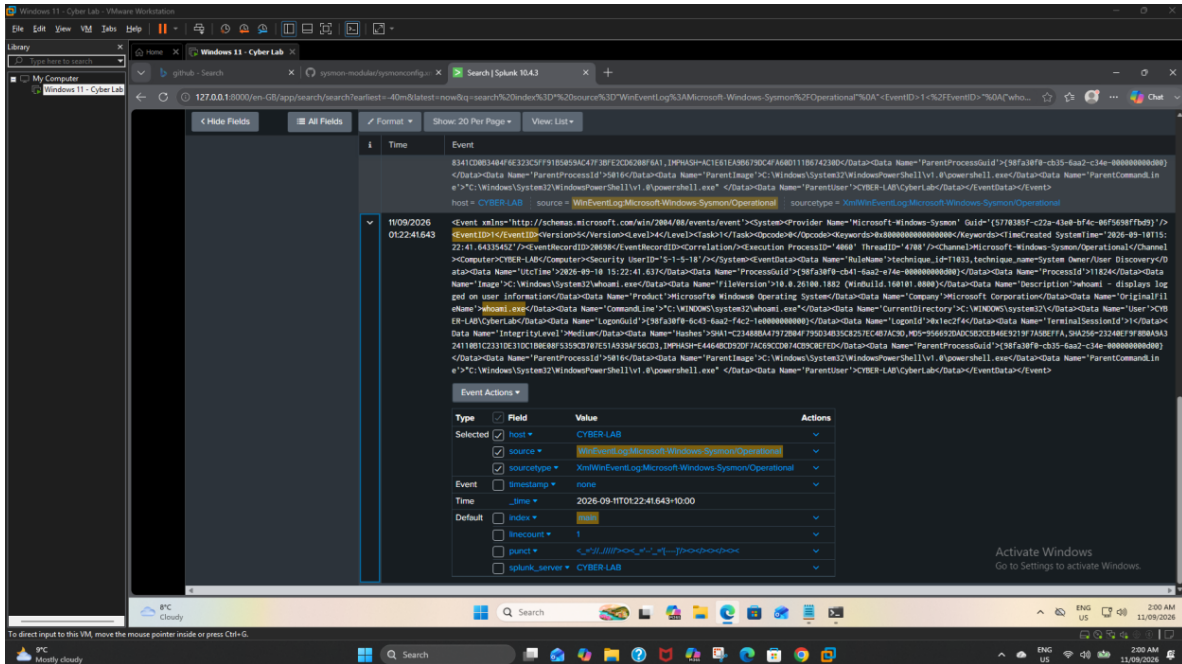


Figure 40. PowerShell PID Correlation.

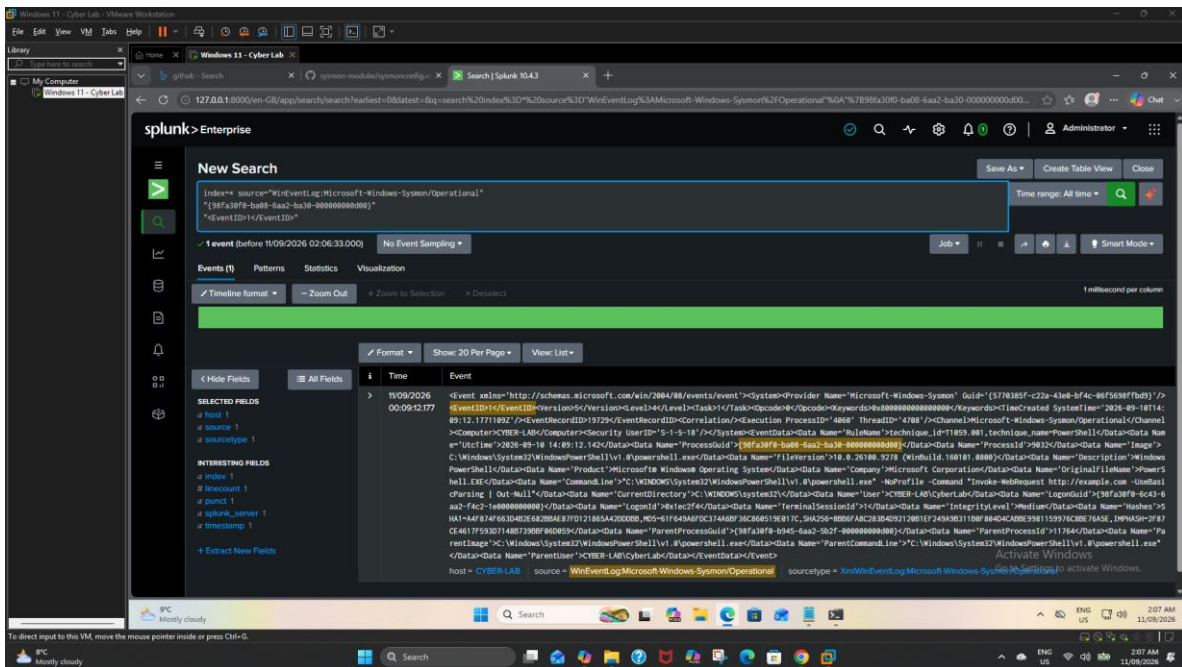


Figure 41. PowerShell ProcessGUID Correlation.

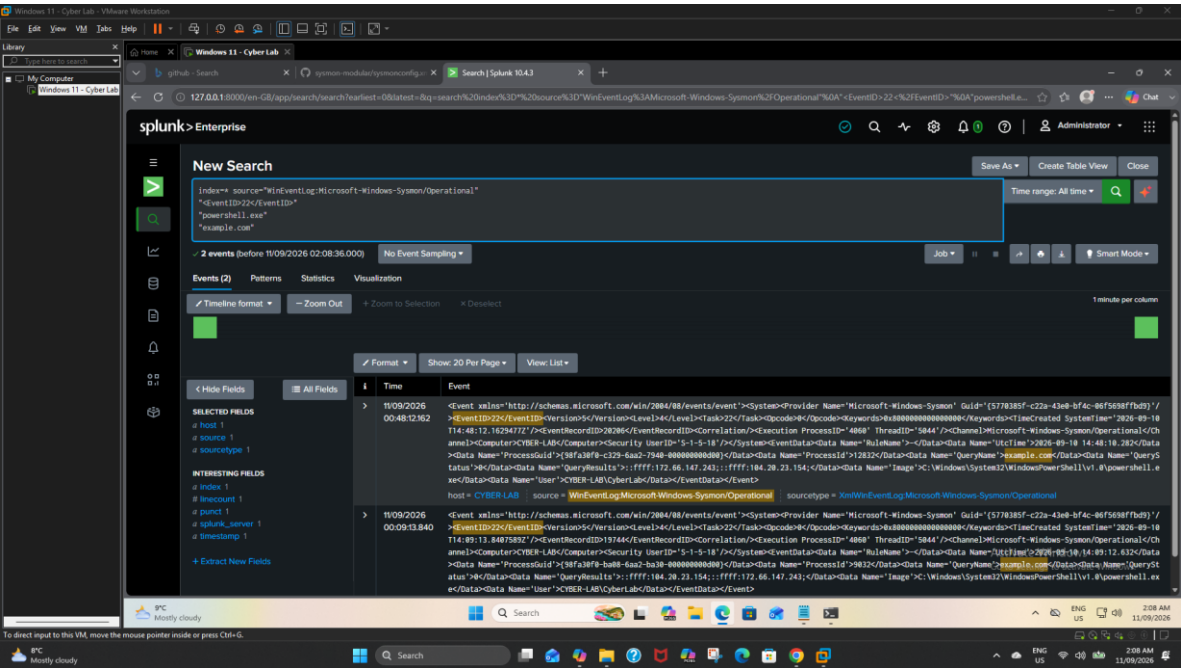


Figure 42. DNS ProcessGuid Correlation.

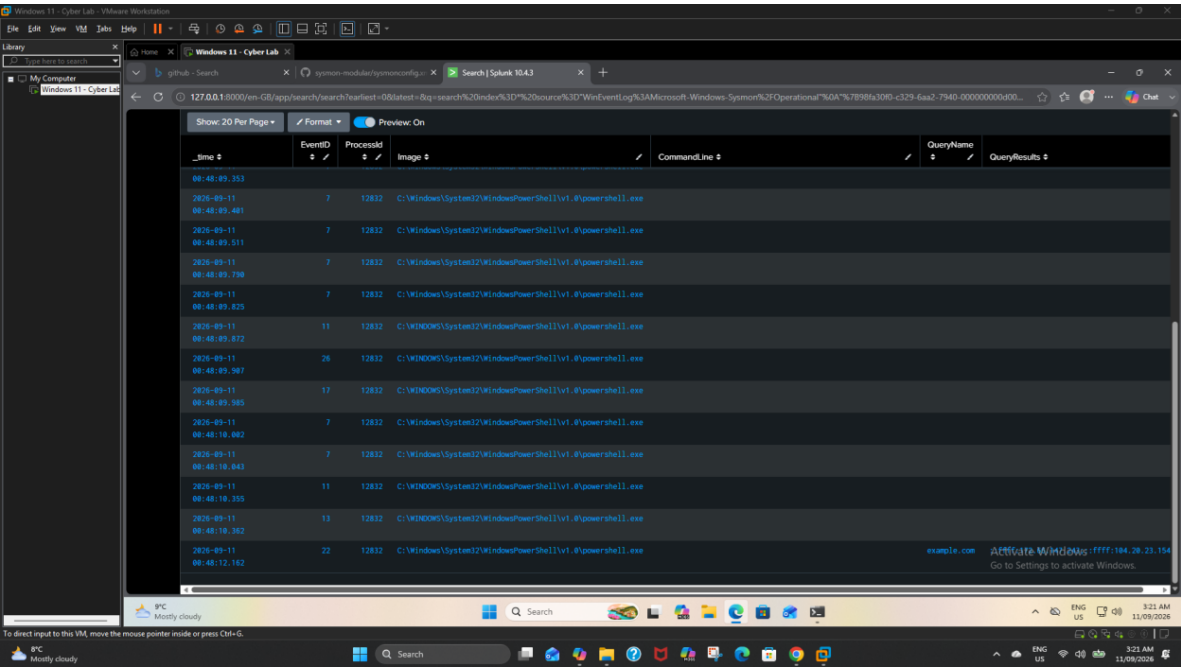


Figure 43. ProcessGuid DNS Correlation.

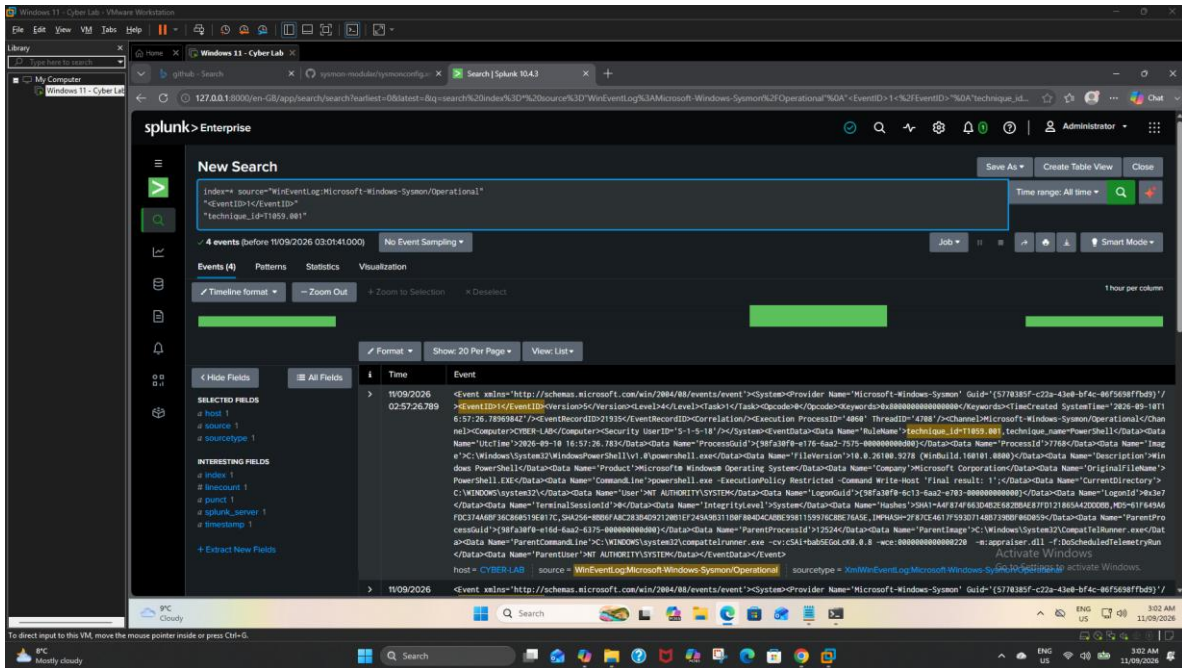


Figure 44. T1059.001 - PowerShell.

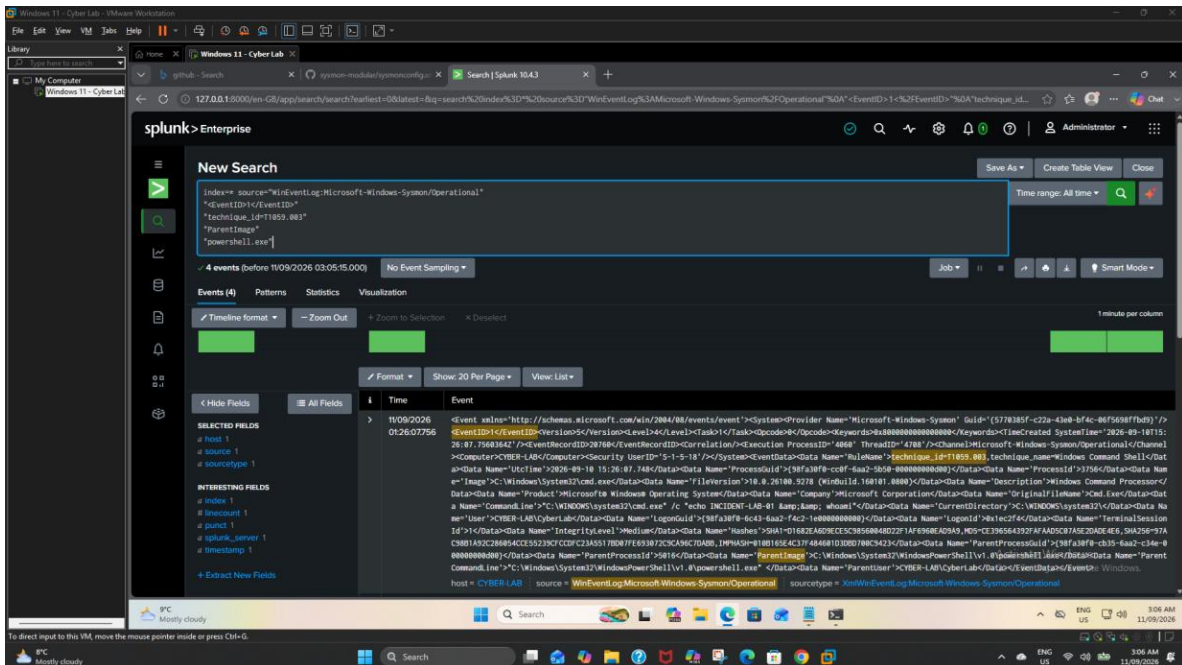


Figure 45. T1059.003 - Windows Command Shell.

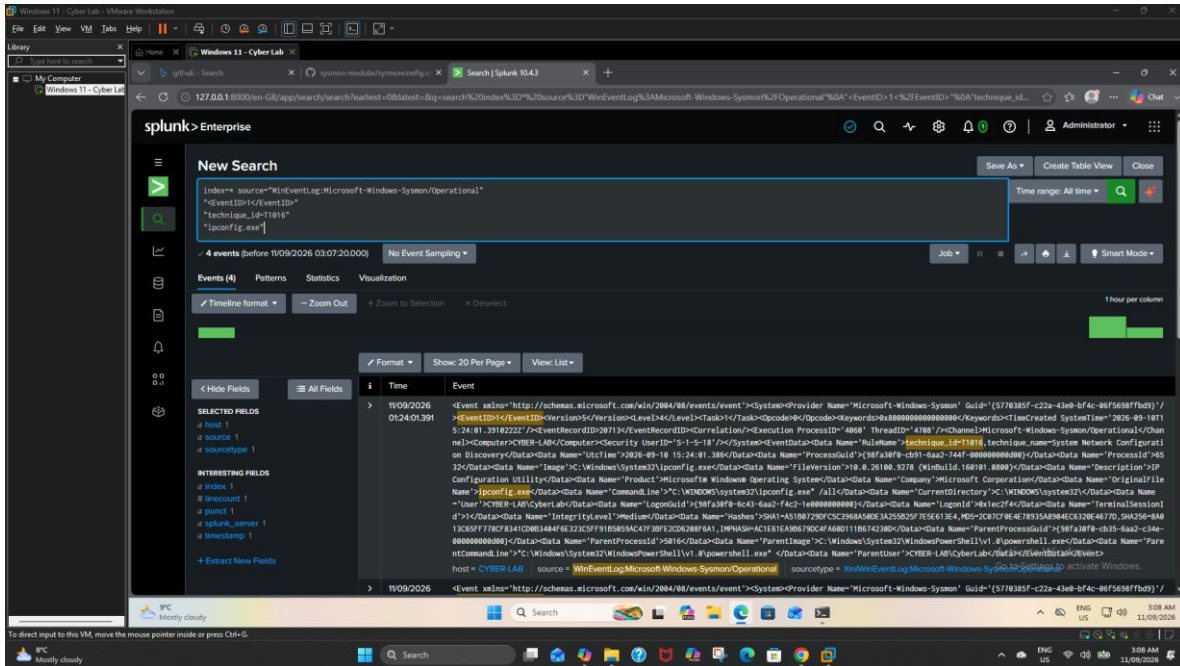


Figure 46. T1016 - System Network Configuration Discovery.

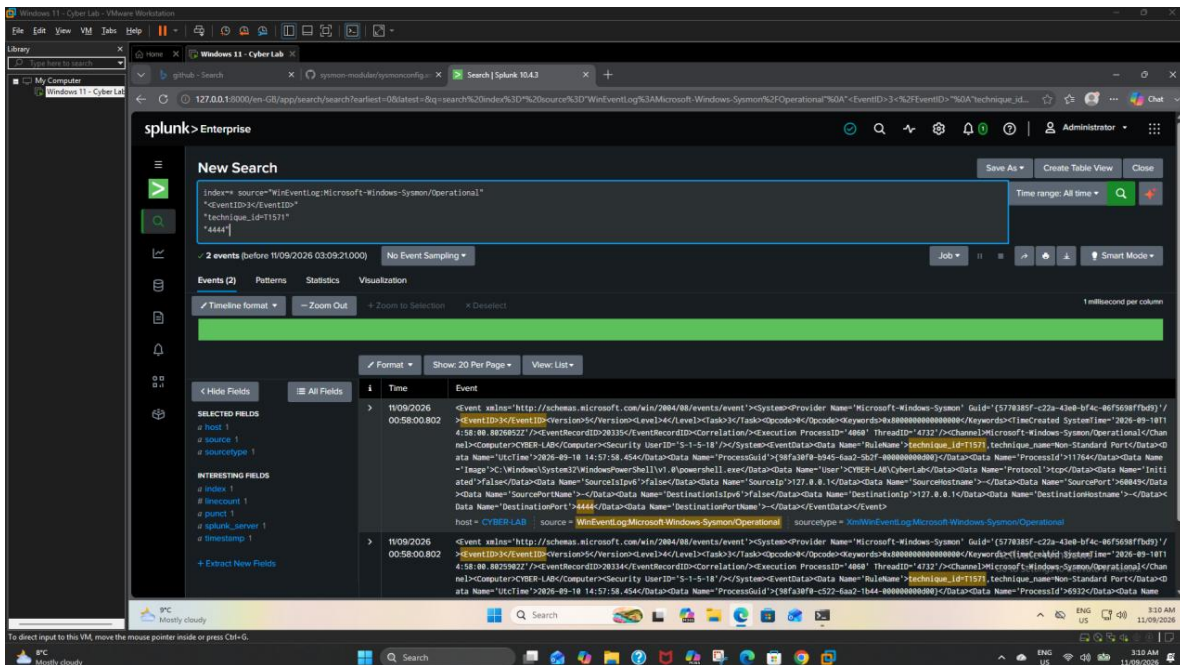


Figure 47. T1571 - Non-Standard Port.

Evidence Handling Note

Screenshots are retained in original form. Overlap is intentional where captures document different stages of validation, detection creation, raw-event inspection and correlation. Benign demonstration infrastructure and loopback traffic are explicitly identified to prevent overstatement of the evidence.